N-body Simulations

by Dimo Dimov

🌀 Multi-Scale N-Body Framework

An Experimental Journey from Planck Scales to Galactic Dynamics

Welcome to a comprehensive computational exploration of **orbital mechanics and gravitational chaos**. This project transitions from the fundamental laws of motion to the complex, unpredictable "dance" of multiple celestial bodies interacting in 1D, 2D, and 3D space.

🔭 Core Objectives

- Simulate stable and chaotic N-body systems.
- Verify Keplerian laws and orbital stability.
- Optimize computation using Barnes-Hut.

🧪 Theoretical Frontiers

- Information-Mass Equivalence Principle.
- Spatially Varying Higgs Field models.
- Quantum boundaries at the Planck Scale.

Created by **Dimo Dimov** | Mathematical Concepts for Developers (2026)

🔴 **IMPORTANT DISCLAIMER: Speculative Content and Potential Inaccuracy**

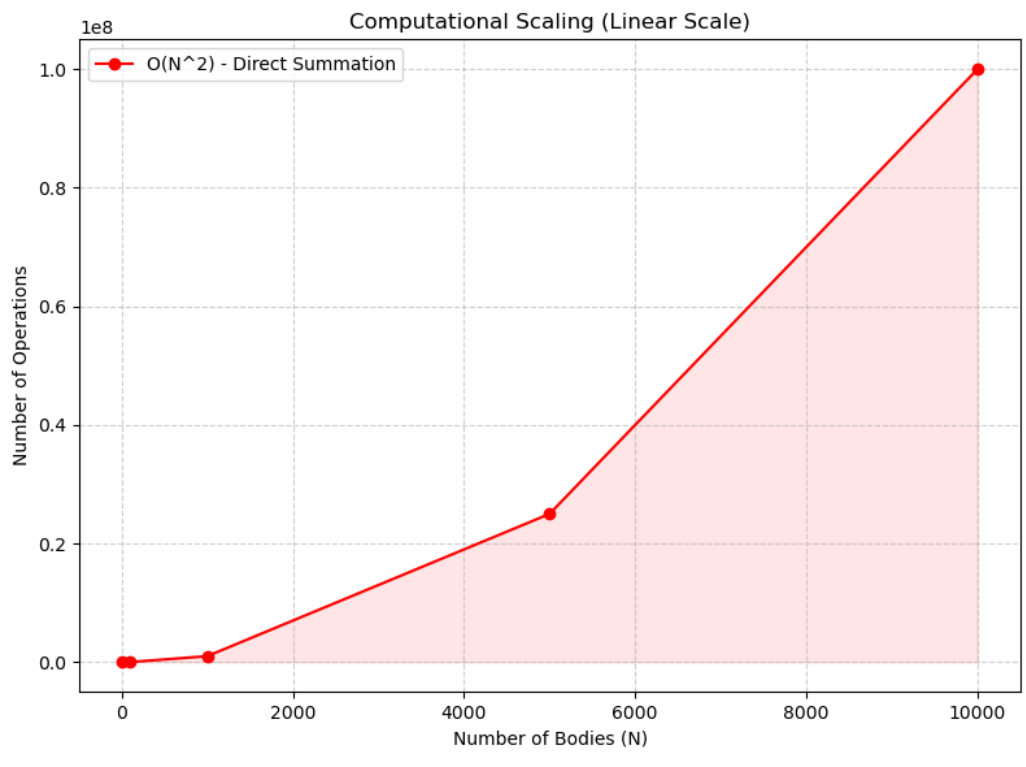
This project is an experimental exploration of N-body systems and **blends established scientific facts with highly speculative models**. Please be advised that the content may contain errors in:

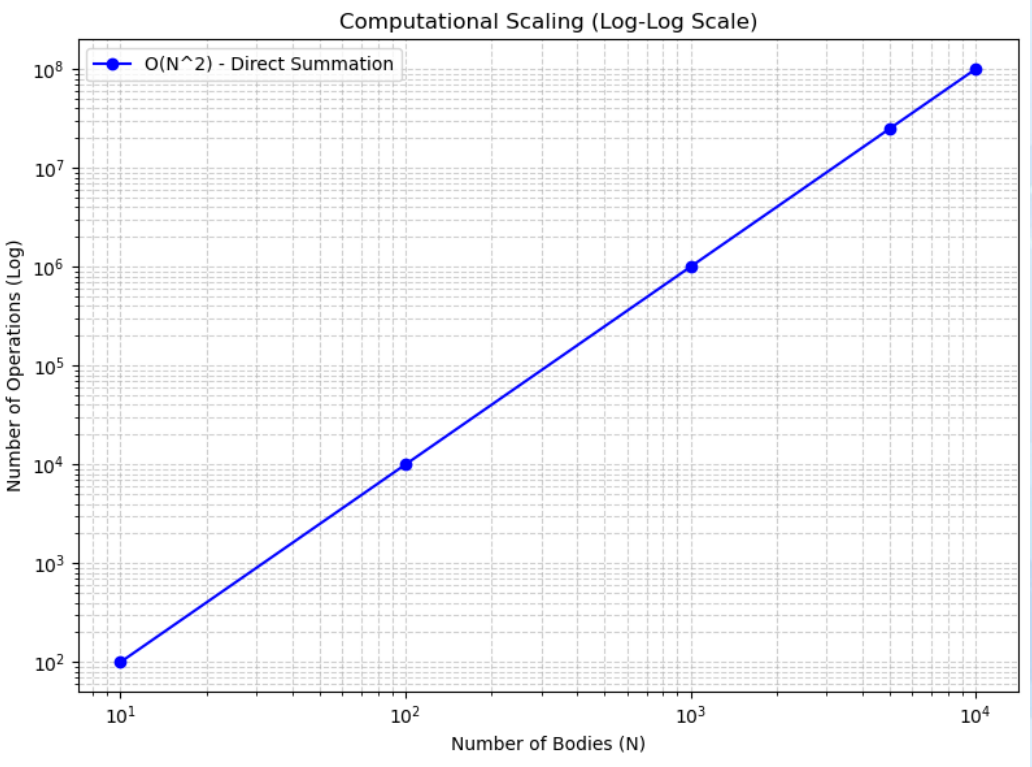
- **Mathematics & Physics:** Formulas and theoretical derivations.
- **Code & Algorithms:** Implementation logic and numerical stability.
- **Simulations & Visualizations:** Accuracy of orbital paths and data representation.

This notebook is intended for educational and creative purposes only. **It must not be taken as absolute scientific truth or used as a reference for professional astronomical calculations.**

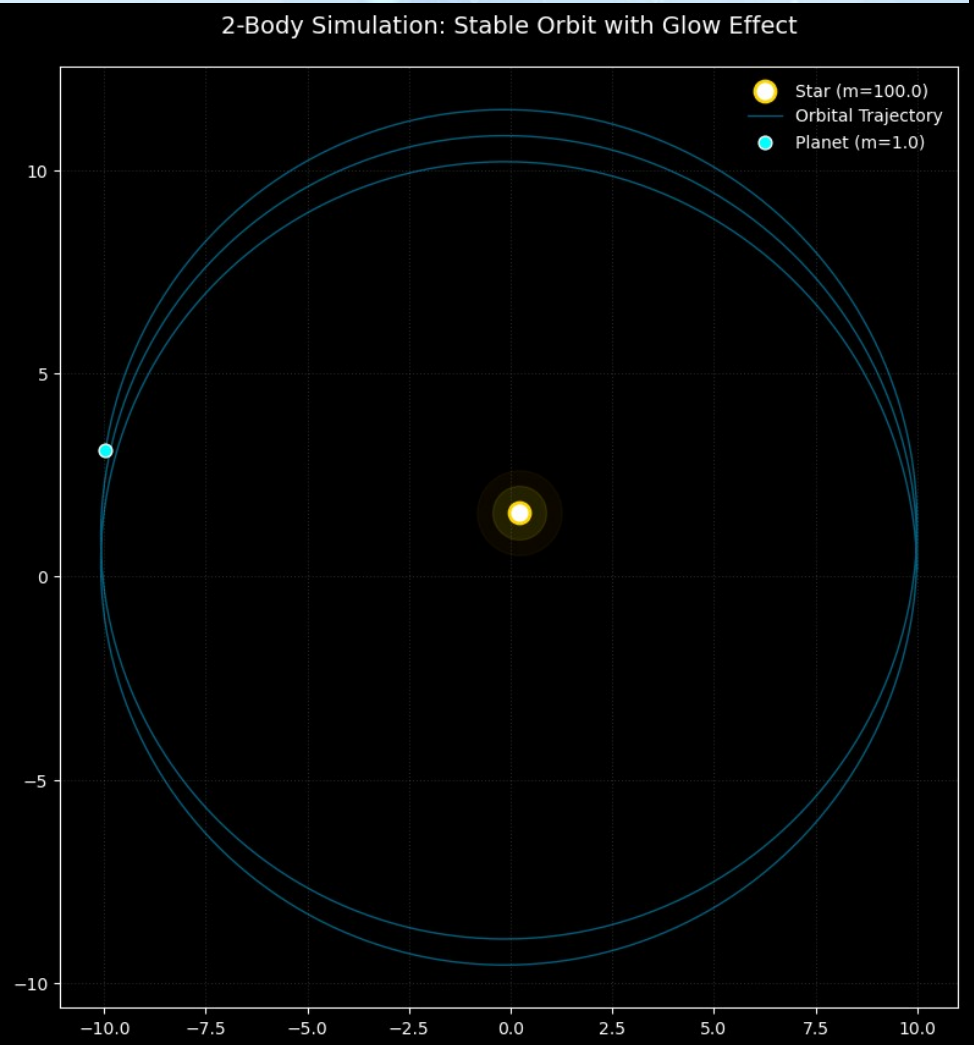
The O(N^2) Barrier

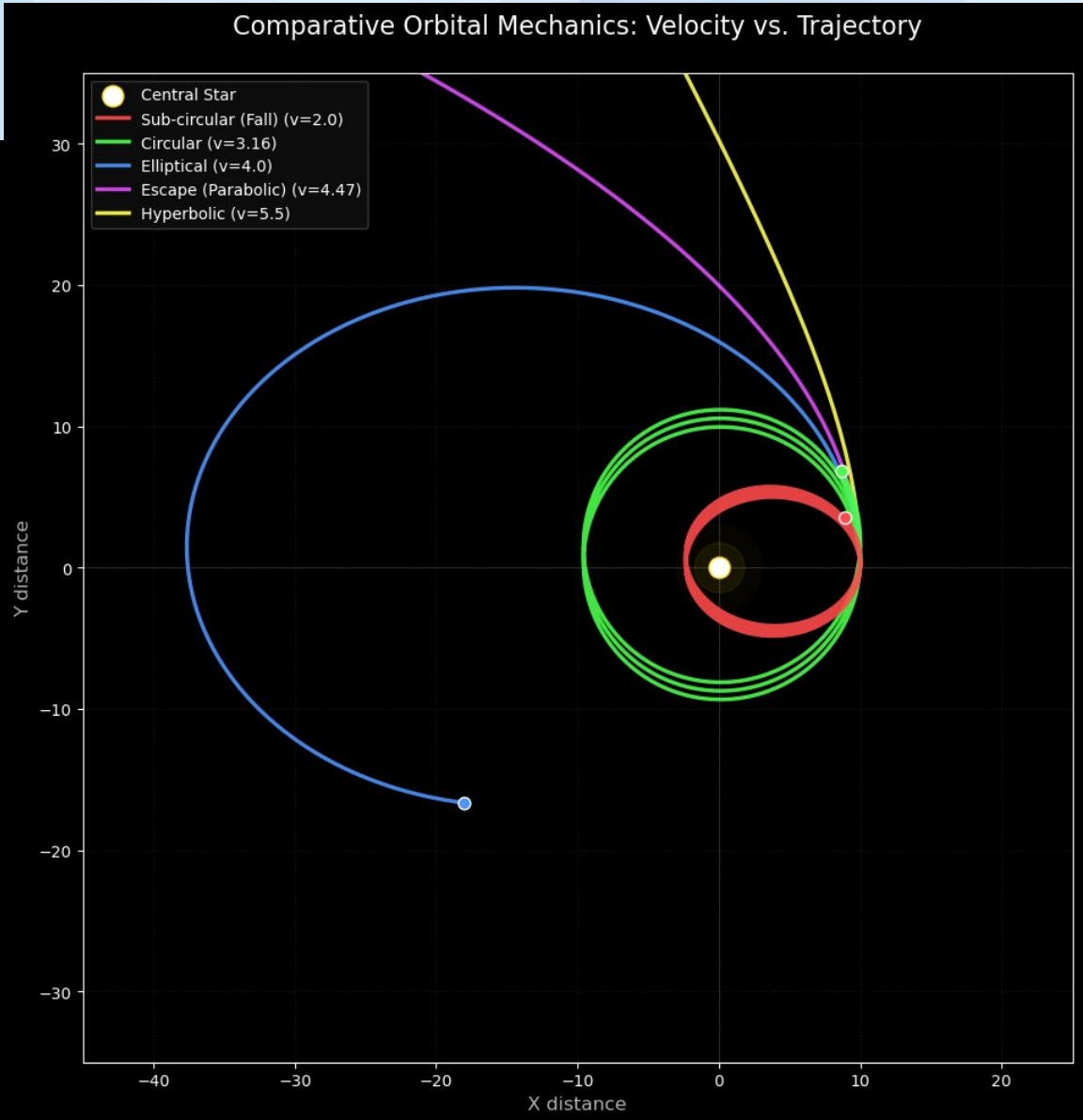
The quadratic scaling O(N^2) of gravitational force calculations renders direct summation computationally prohibitive for large-scale systems.



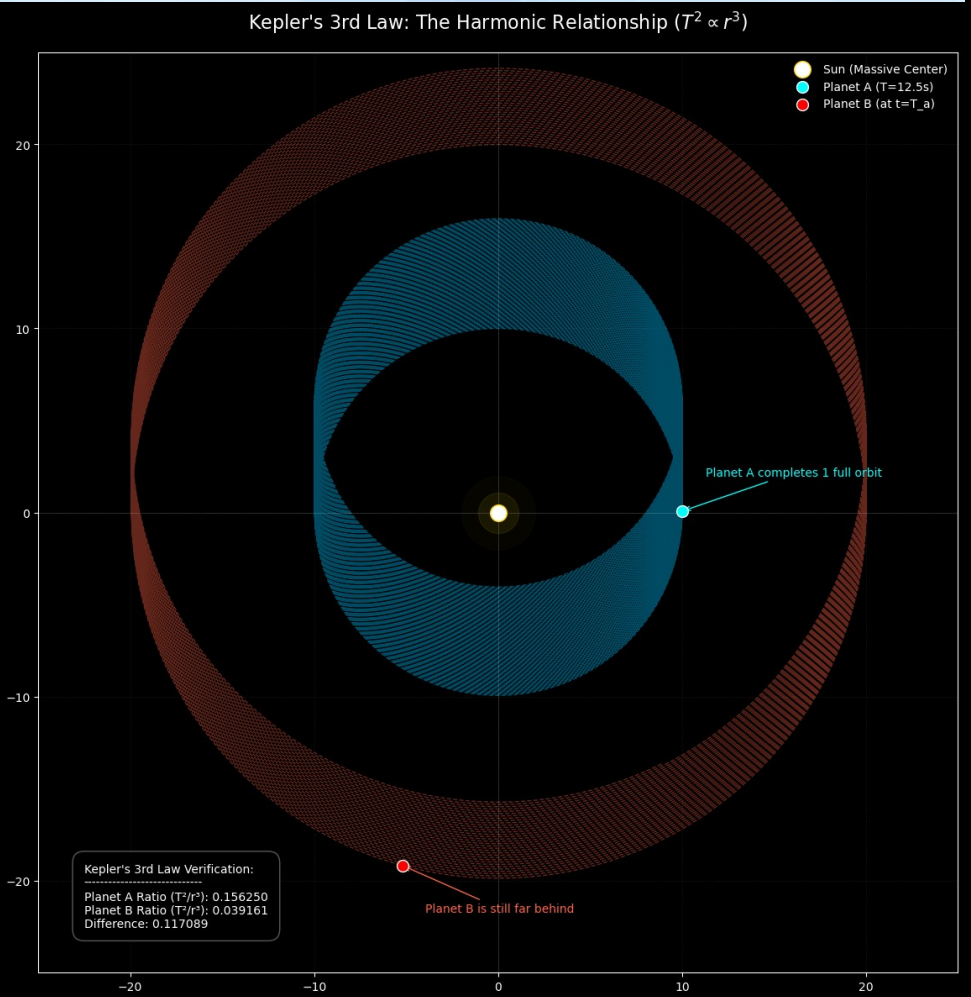


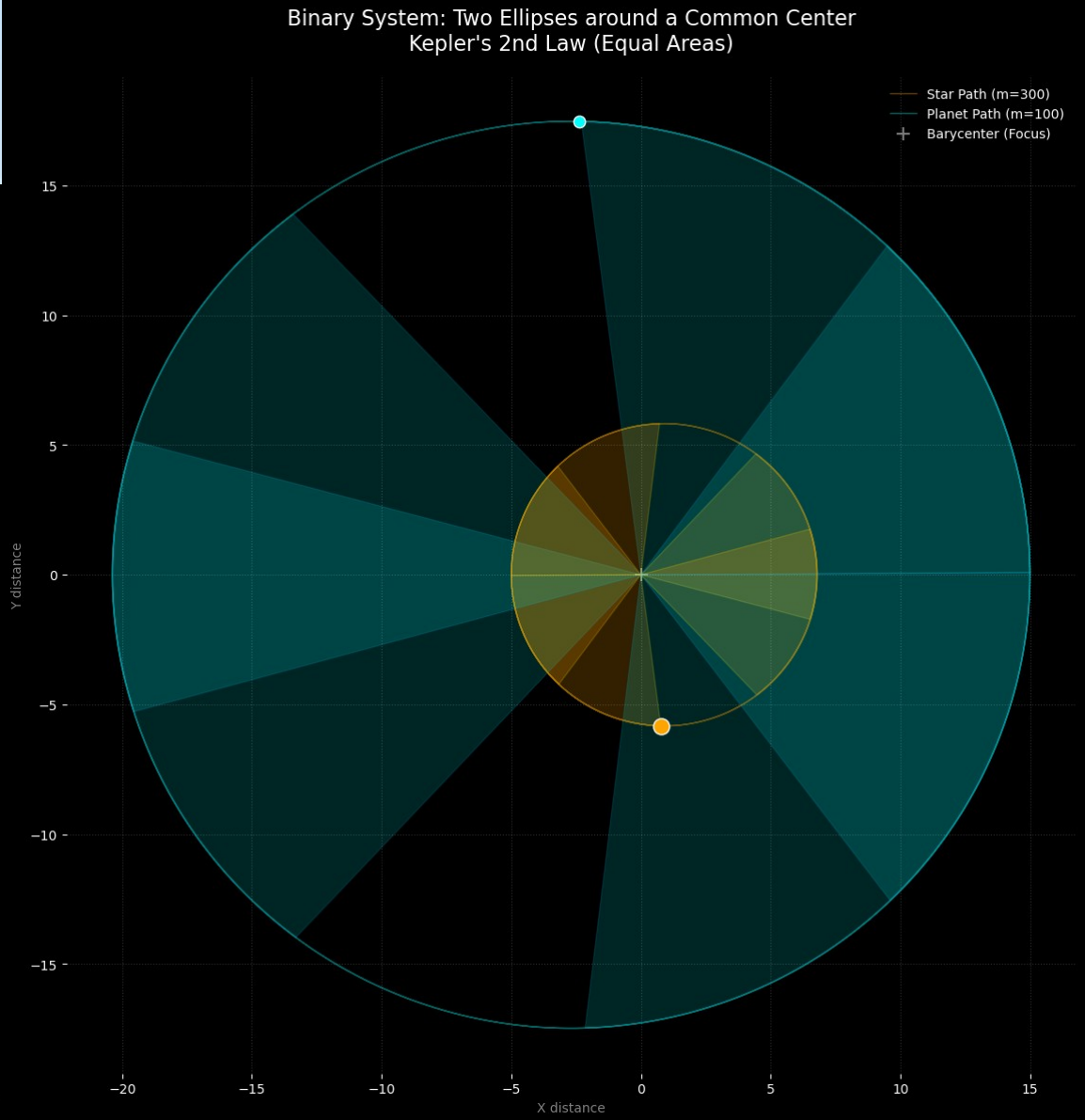
Orbital Mechanics





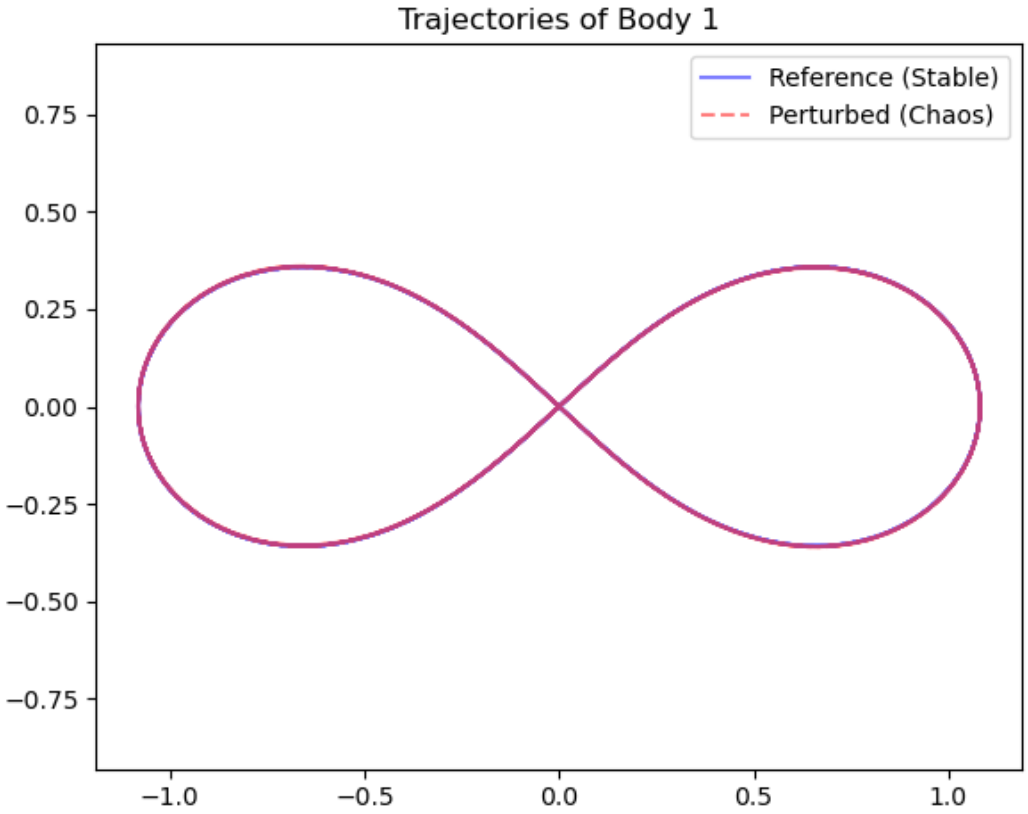
Kepler's Laws

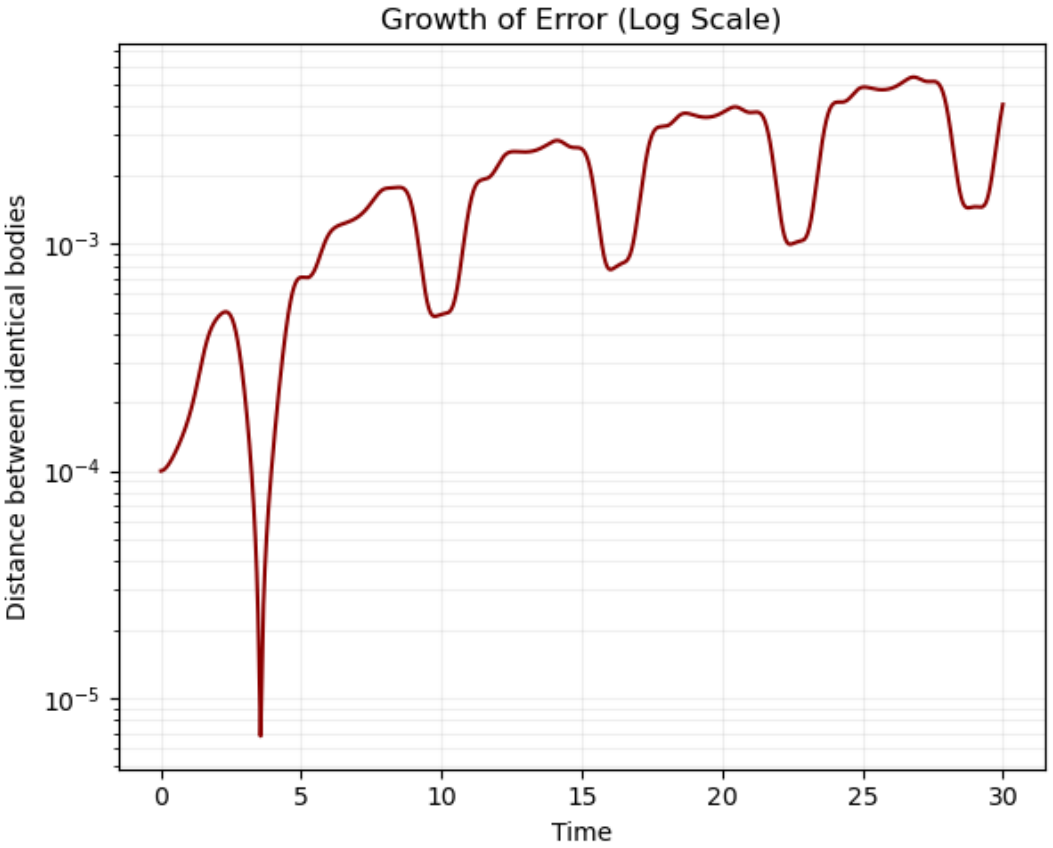


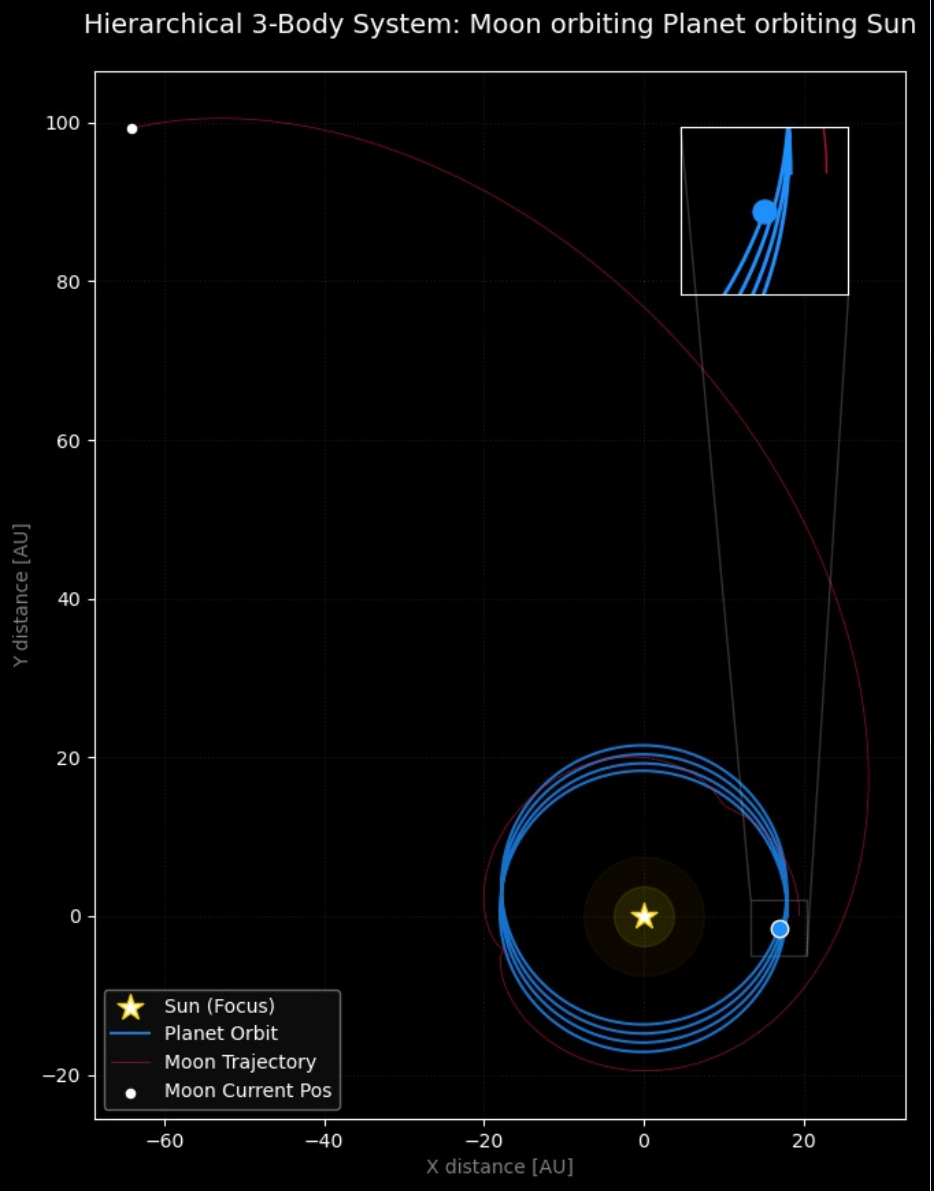


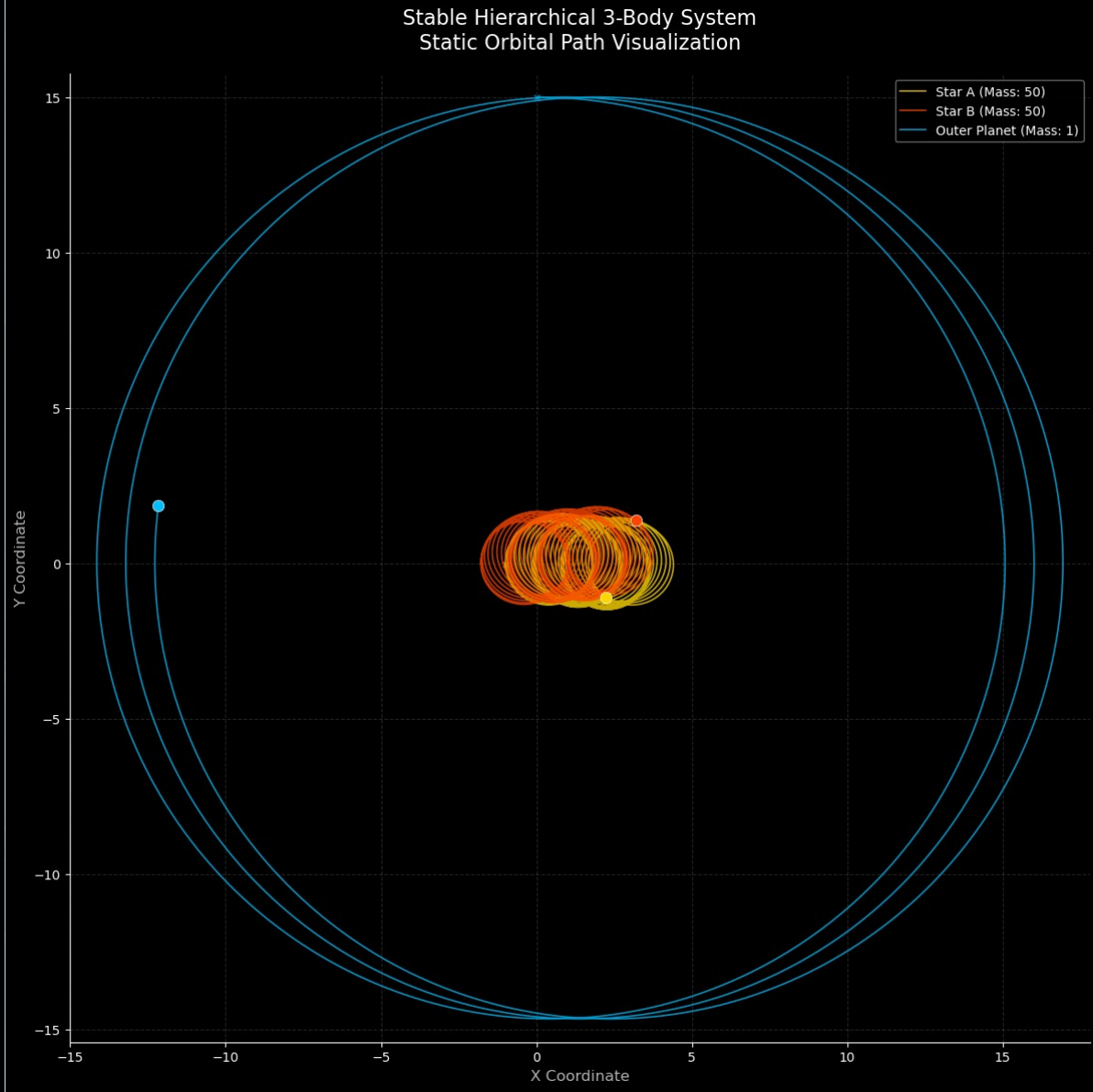
Extraordinary "Figure-Eight" Orbit

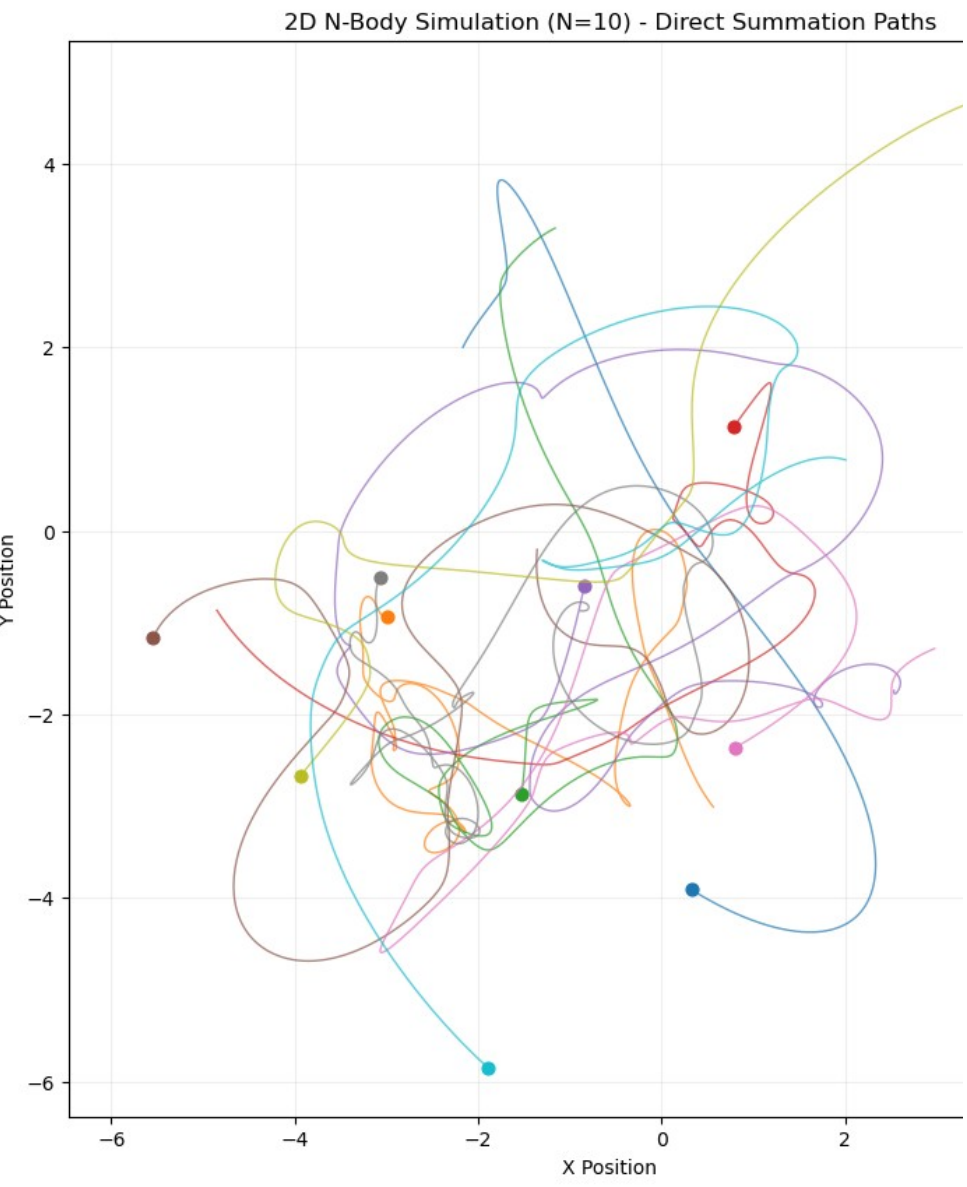
While most 3-body systems are chaotic, a few periodic solutions exist.

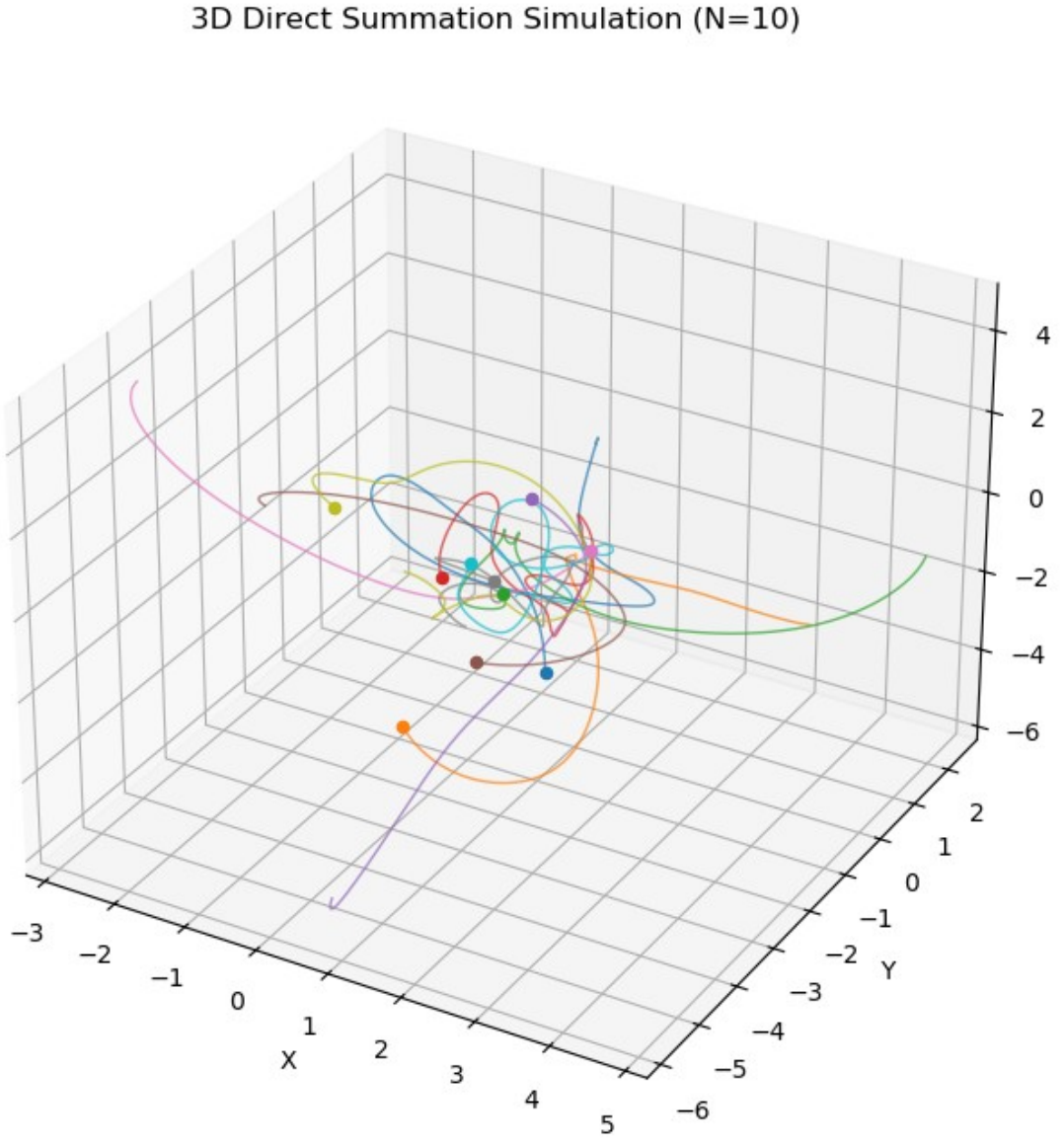





Hierarchical 3-Body System: Moon orbiting Planet orbiting Sun
Y distance [AU]
X distance [AU]
Sun (Focus)
Planet Orbit
Moon Trajectory
Moon Current Pos

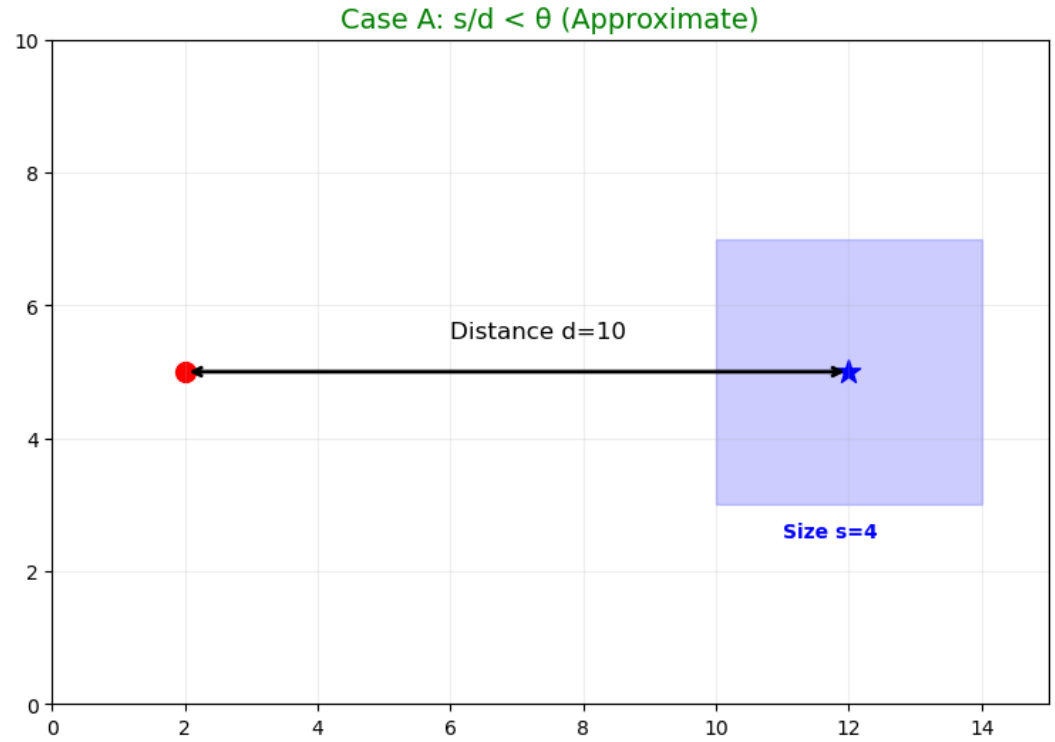

Stable Hierarchical 3-Body System
Static Orbital Path Visualization
Star A (Mass: 50)
Star B (Mass: 50)
Outer Planet (Mass: 1)
Y Coordinate
X Coordinate

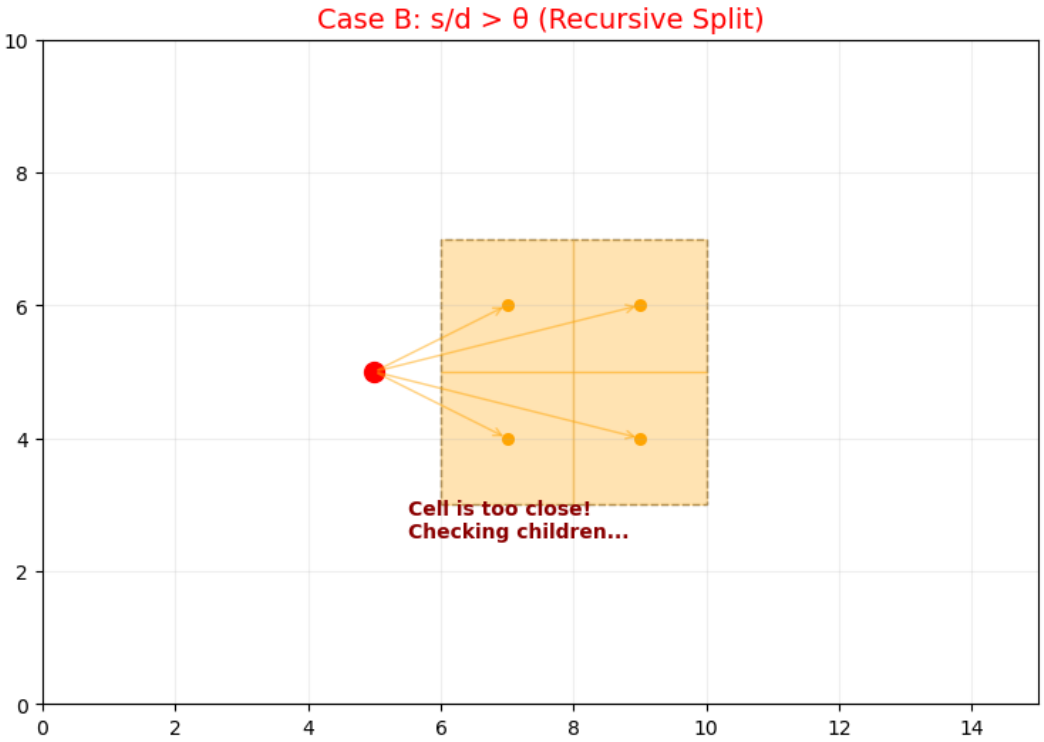

2D N-Body Simulation (N=10) - Direct Summation Paths
Y Position
X Position
4
2
0
−2
−4
−6
−6
−4
−2
0
2

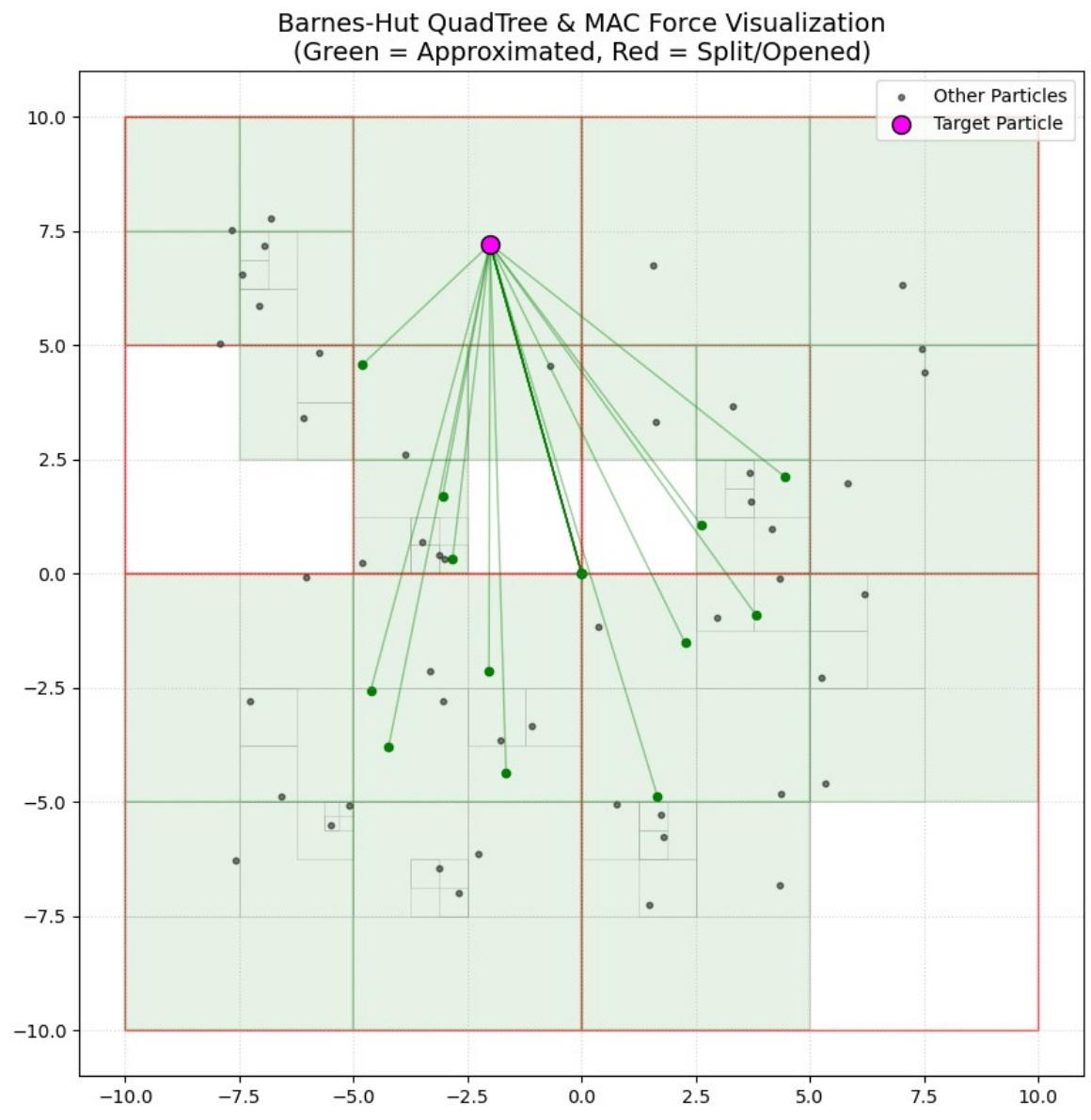

3D Direct Summation Simulation (N=10)
X
Y
−3
−2
−1
0
1
2
3
4
5
−6
−5
−4
−3
−2
−1
0
1
2
4
2
0
−2
−4
−6

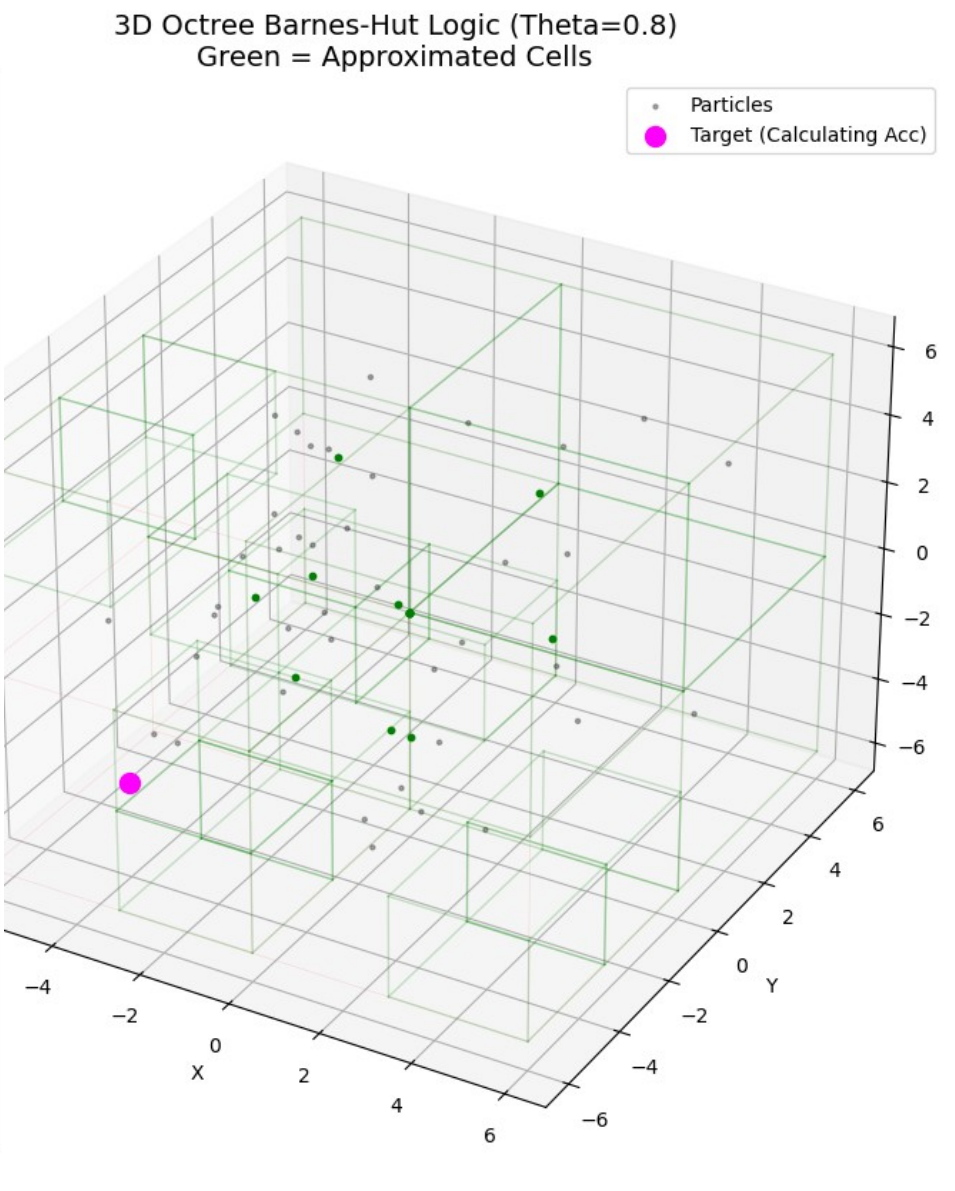
Barnes-Hut Approximation

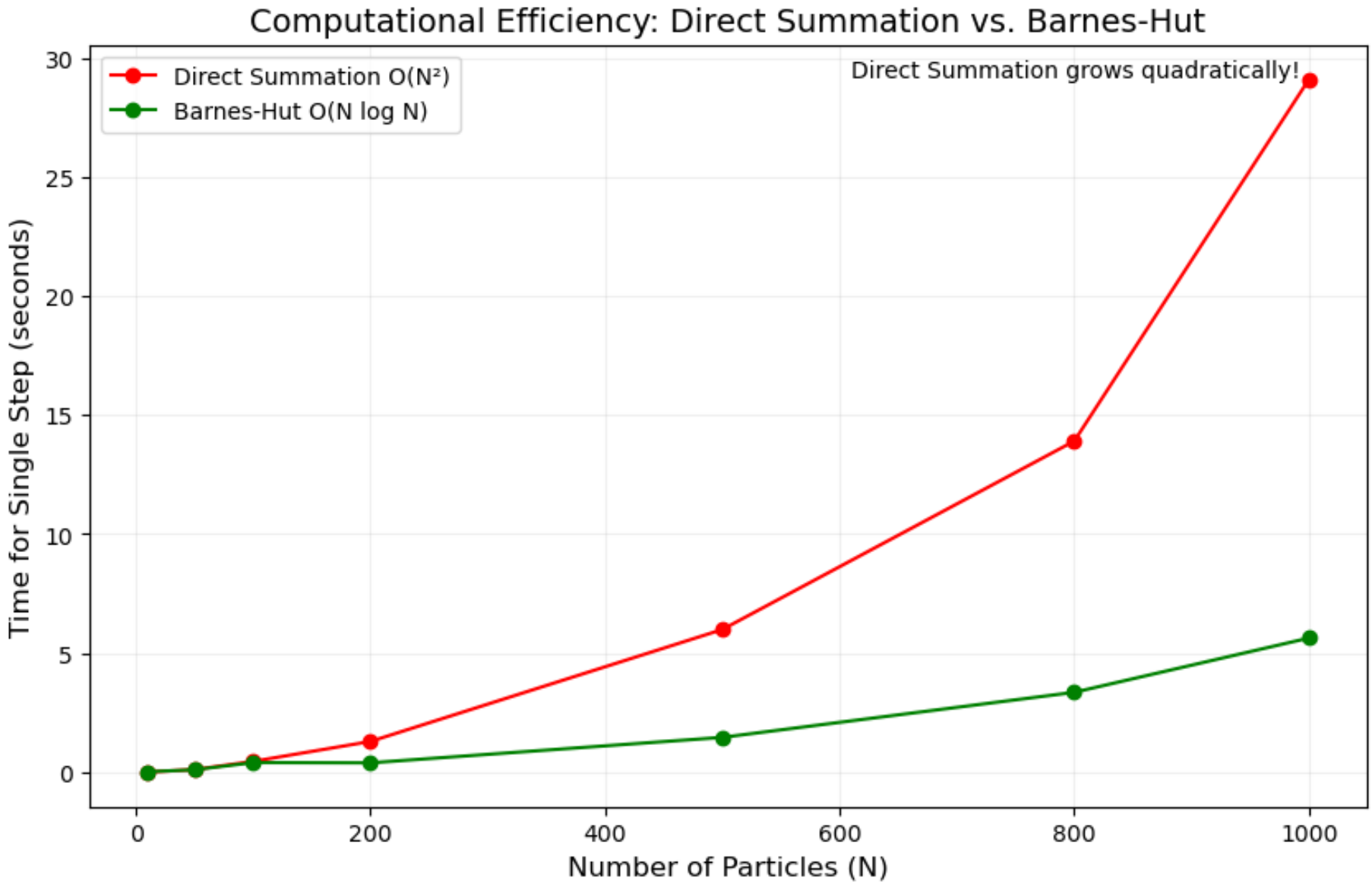
It relies on the observation that the gravitational force from a distant cluster of particles can be approximatedas the force from a single "super-particle" located at the cluster's center of mass.





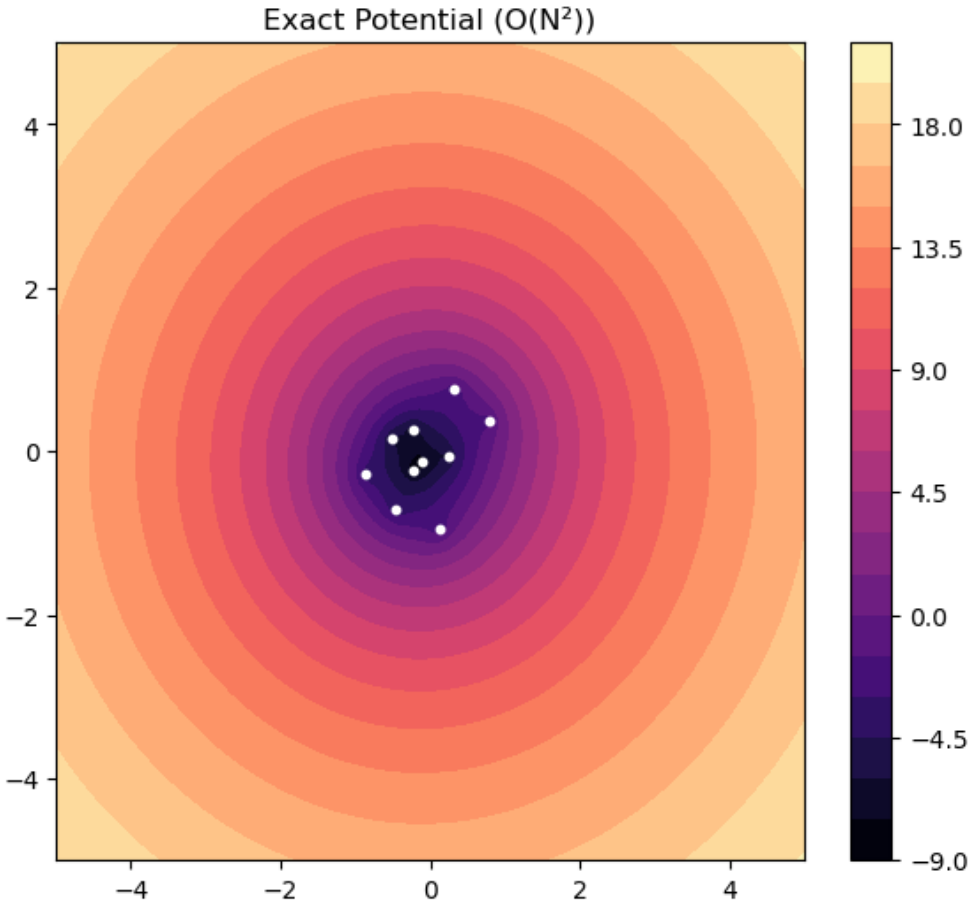

Barnes-Hut QuadTree & MAC Force Visualization
(Green = Approximated, Red = Split/Opened)
Other Particles
Target Particle
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0

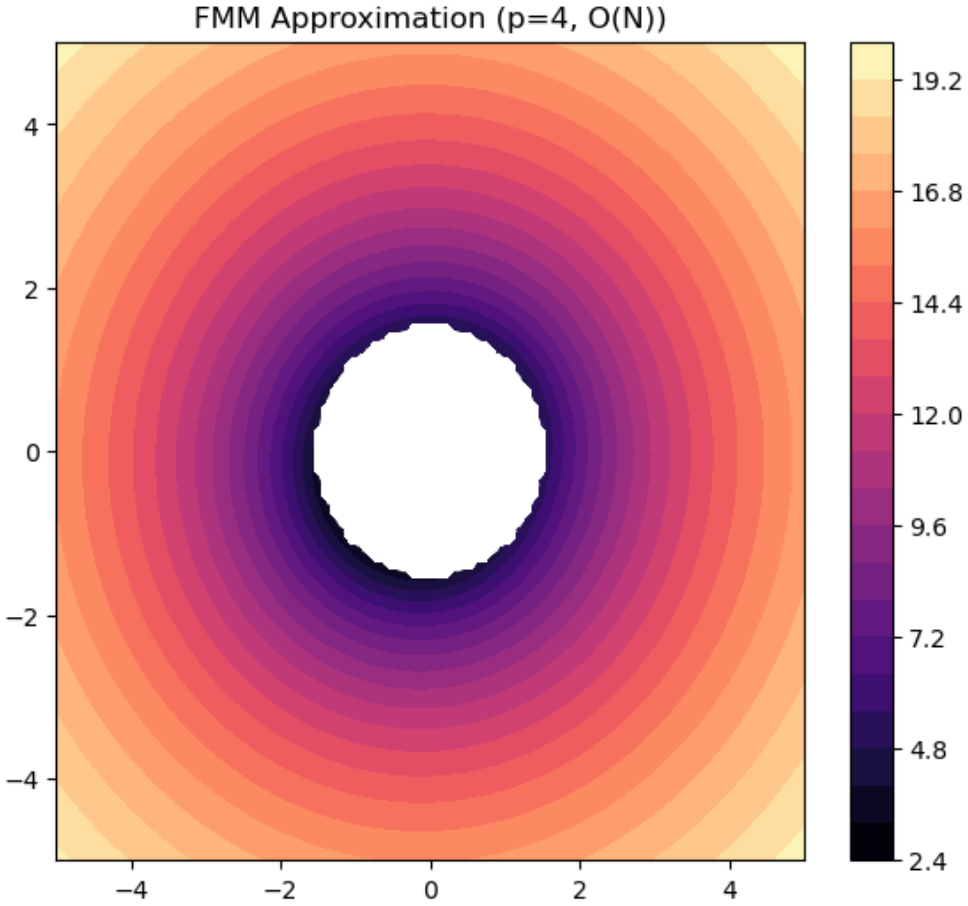

3D Octree Barnes-Hut Logic (Theta=0.8)
Green = Approximated Cells
Particles
Target (Calculating Acc)
X
Y
−4
−2
0
2
4
6

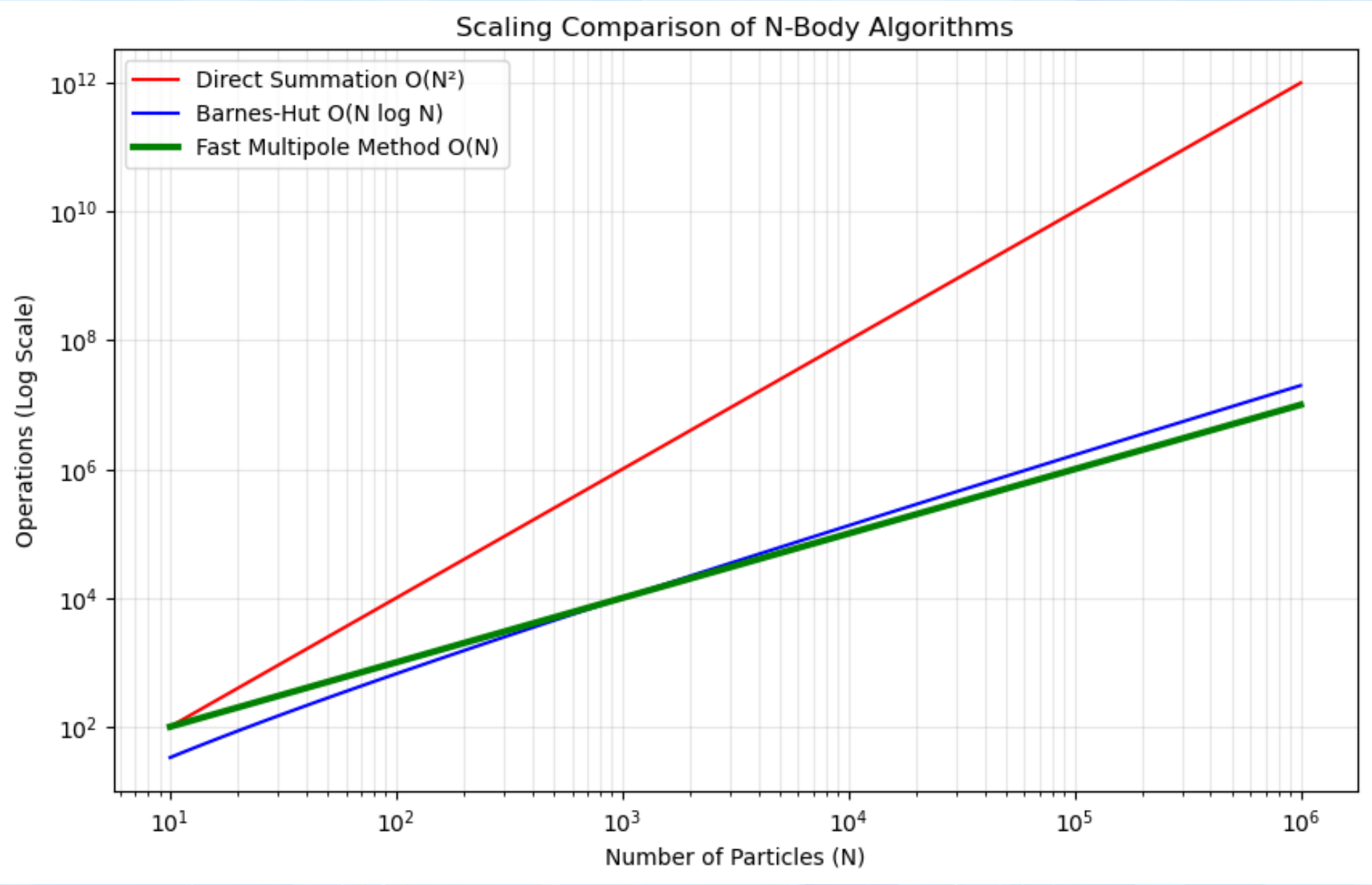

Computational Efficiency: Direct Summation vs. Barnes-Hut
Direct Summation O(N²)
Barnes-Hut O(N log N)
Direct Summation grows quadratically!
Time for Single Step (seconds)
Number of Particles (N)
0
5
10
15
20
25
30
0
200
400
600
800
1000

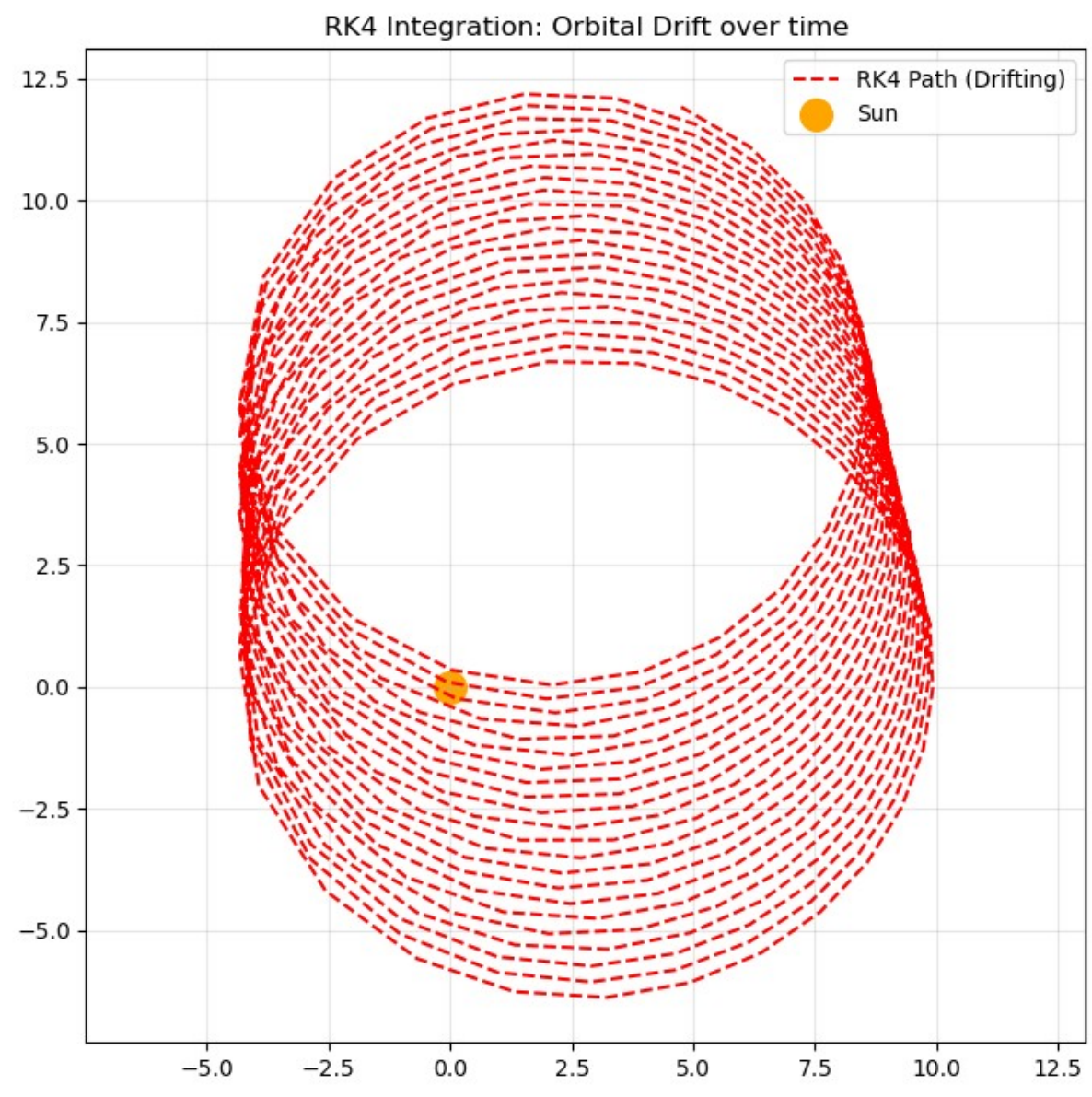
Algorithmic Peak: Fast Multipole Method

We represent the entire gravitational potential of a cell using a Multipole Expansion.






Scaling Comparison of N-Body Algorithms
Direct Summation O(N²)
Barnes-Hut O(N log N)
Fast Multipole Method O(N)
Operations (Log Scale)
Number of Particles (N)
10^{12}
10^{10}
10^{8}
10^{6}
10^{4}
10^{2}
10^{1}
10^{2}
10^{3}
10^{4}
10^{5}
10^{6}



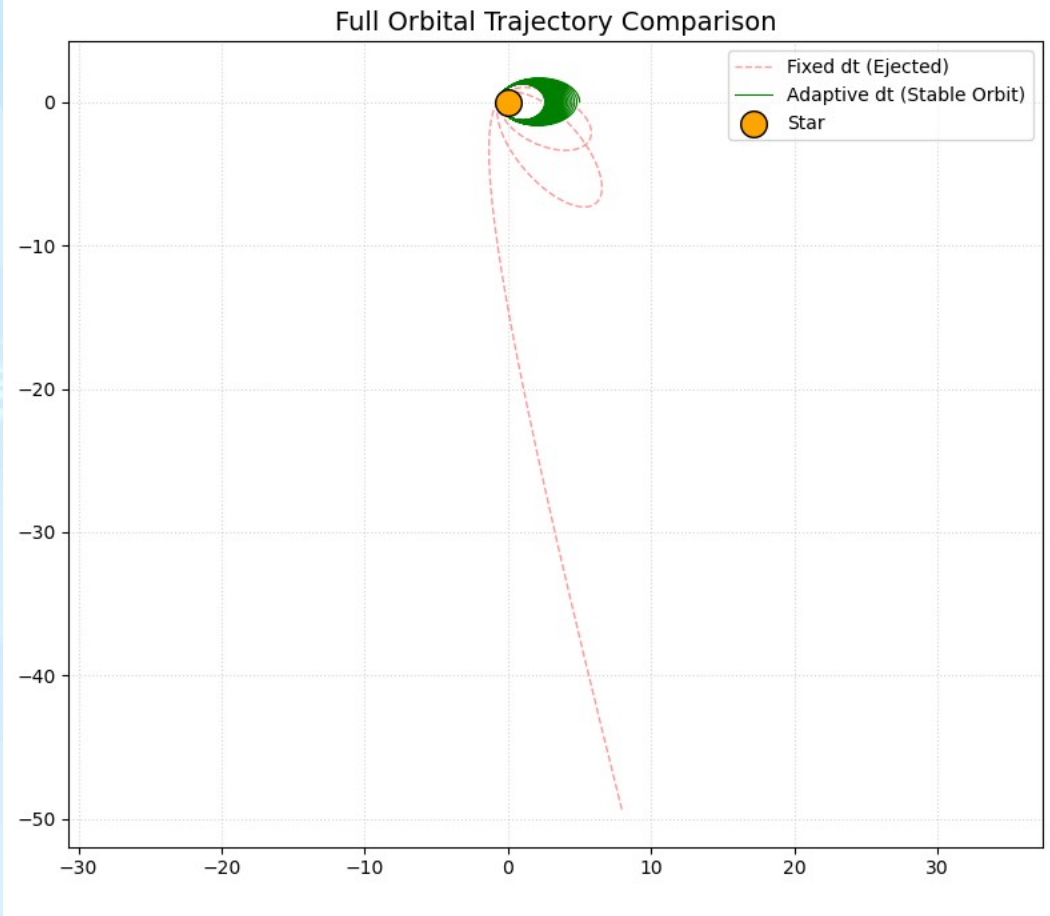
Numerical Integration: Runge-Kutta (RK4)

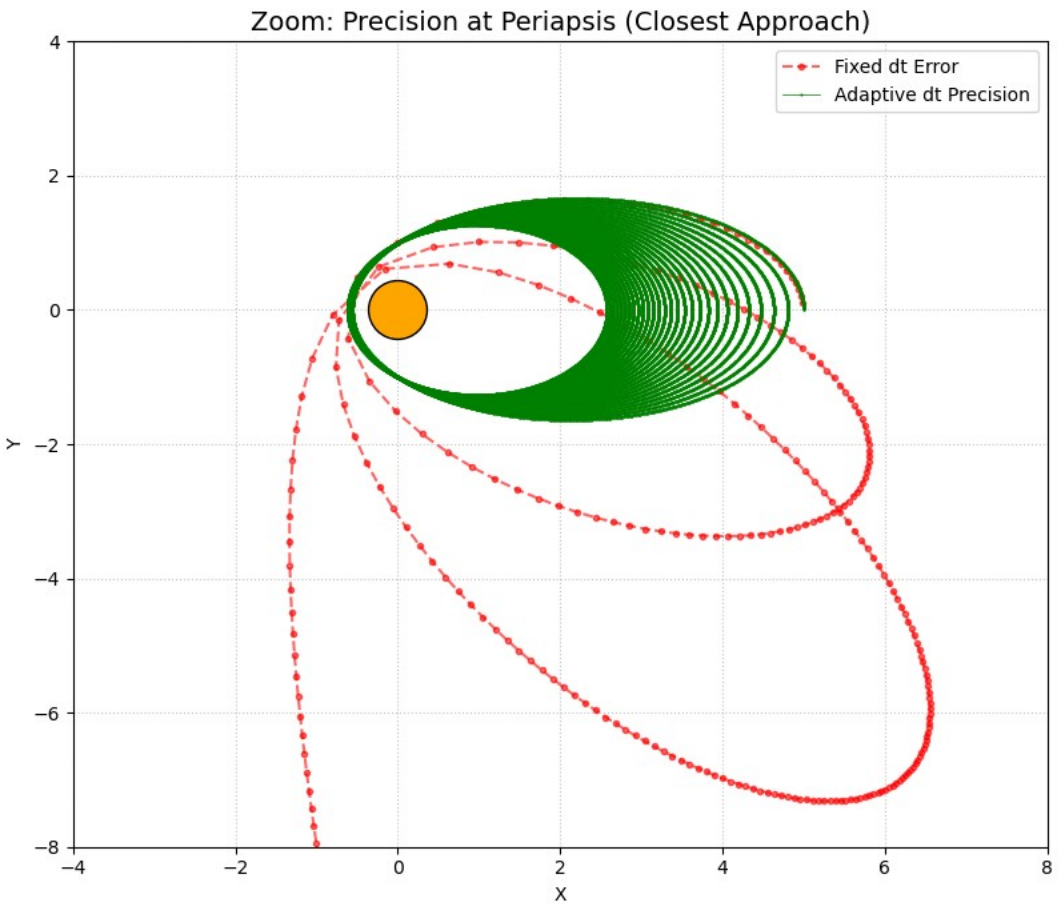
The RK4 method approximates the next state of a system by taking a weighted average of four increments (slopes) within a single time step **Δt**.

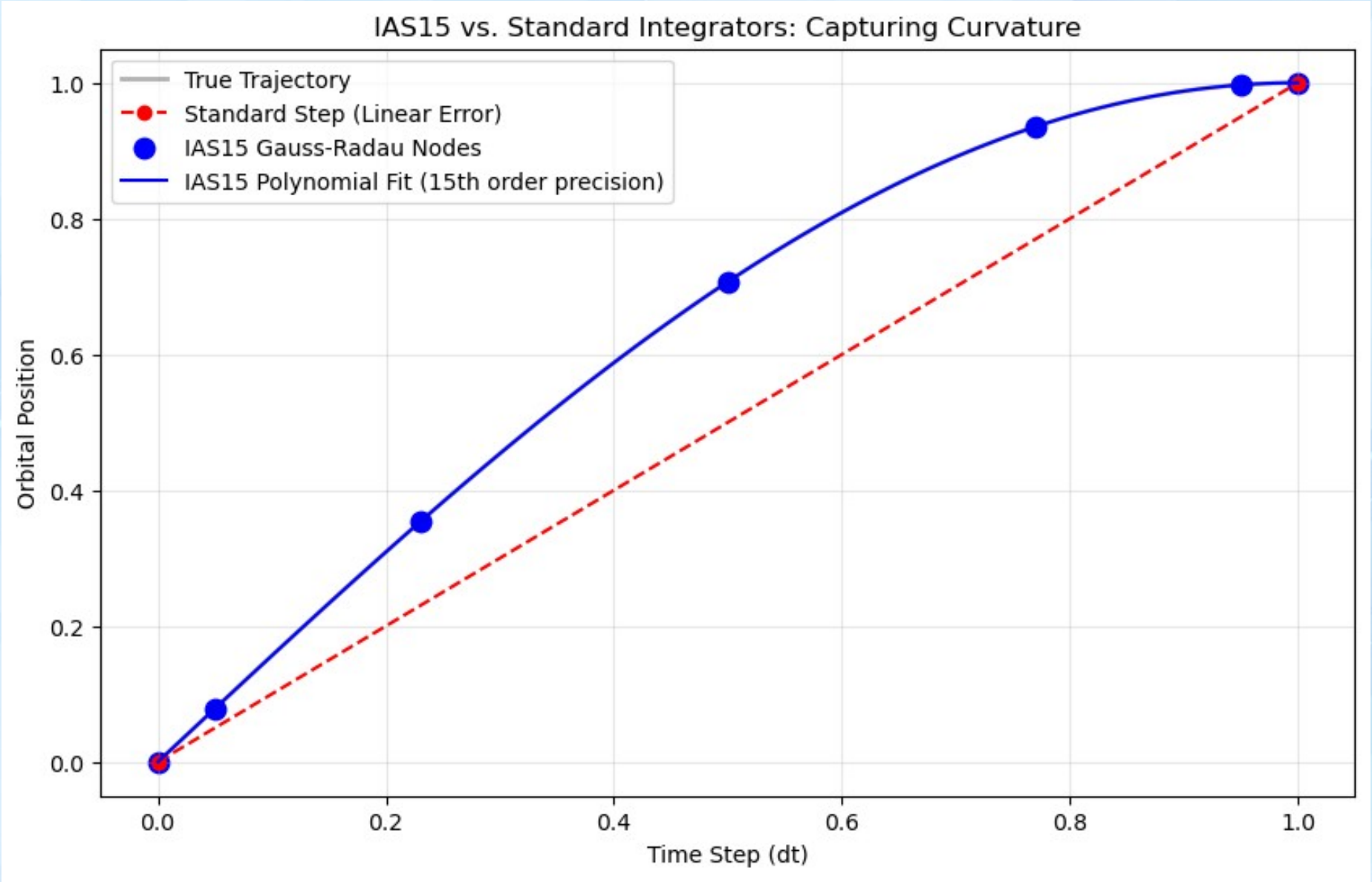
Why it "drifts": Unlike Symplectic integrators, RK4 does not preserve the phase-space volume.

Velocity Verlet

The Velocity Verlet algorithm is a symplectic integrator.

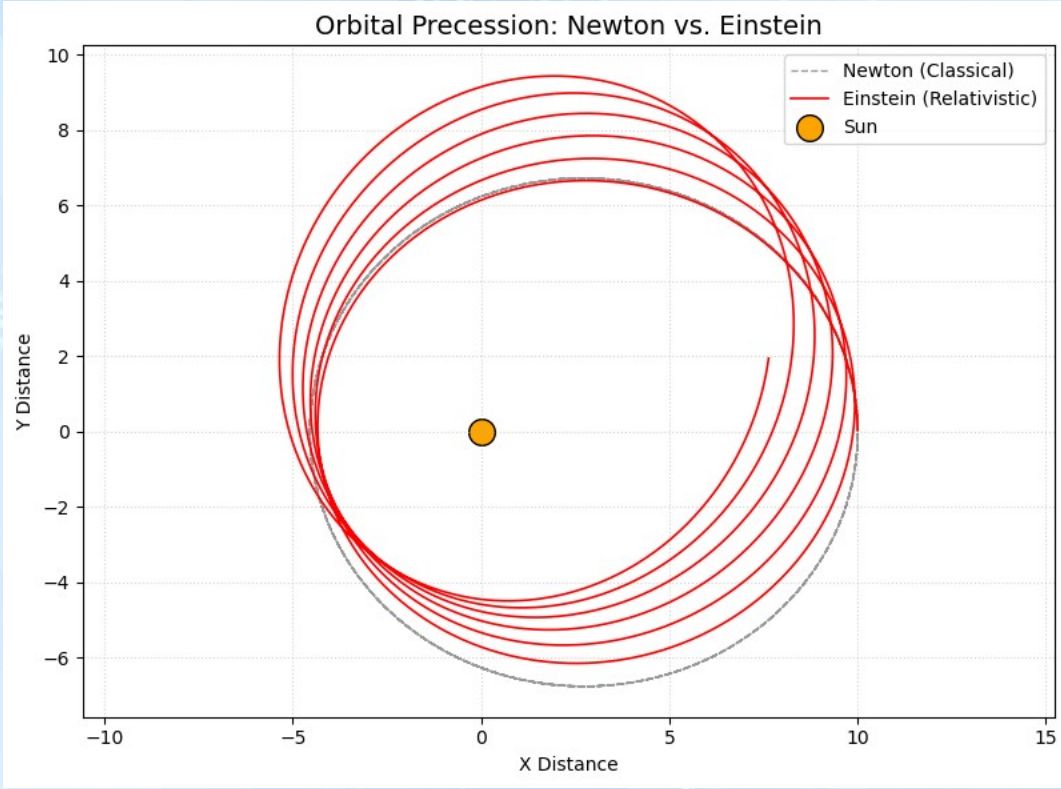


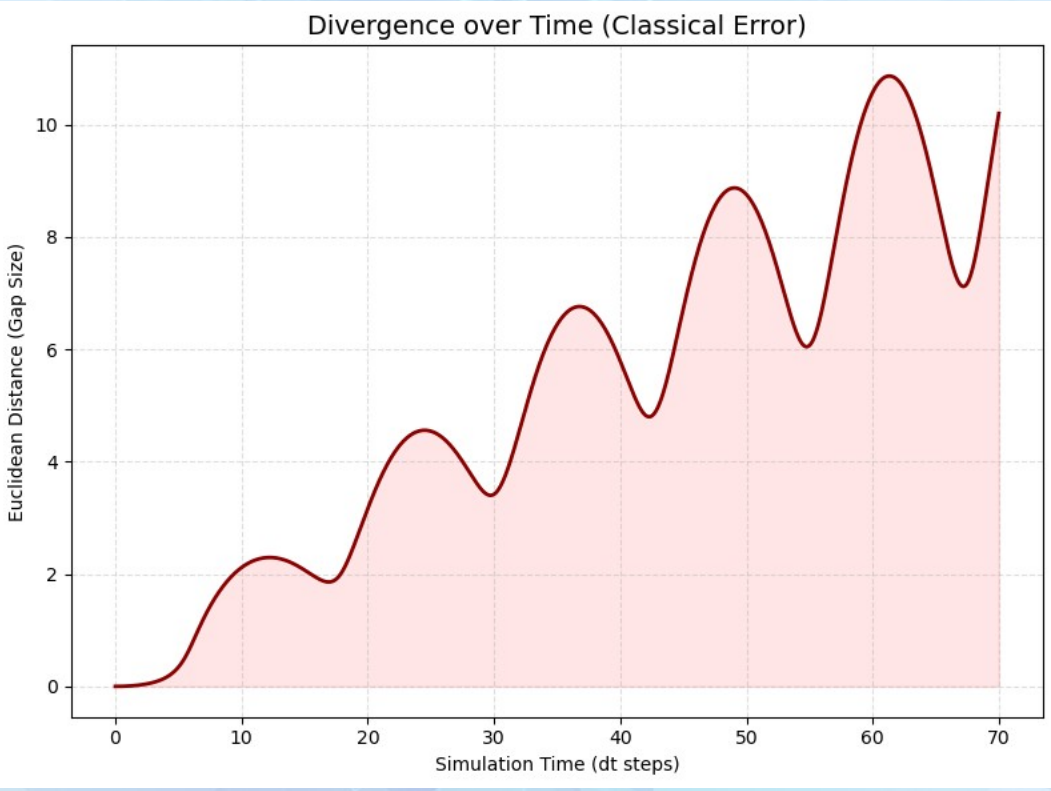


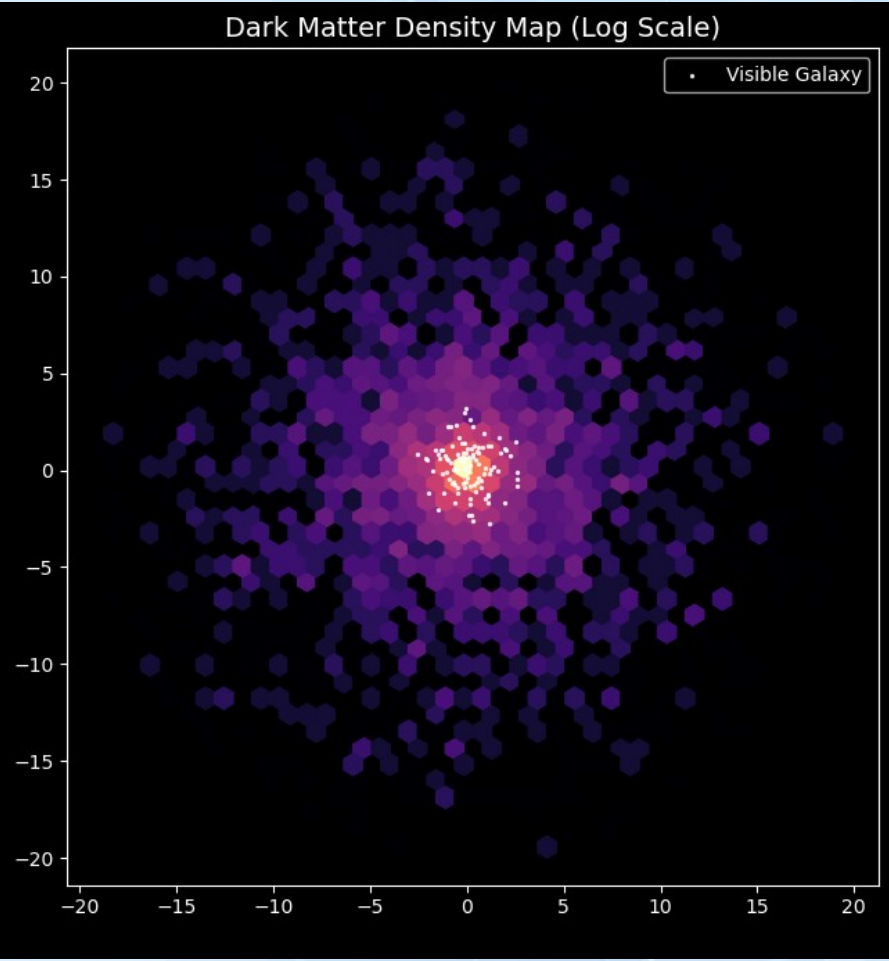

IAS15 vs. Standard Integrators: Capturing Curvature
True Trajectory
Standard Step (Linear Error)
IAS15 Gauss-Radau Nodes
IAS15 Polynomial Fit (15th order precision)
Orbital Position
Time Step (dt)
0.0
0.2
0.4
0.6
0.8
1.0

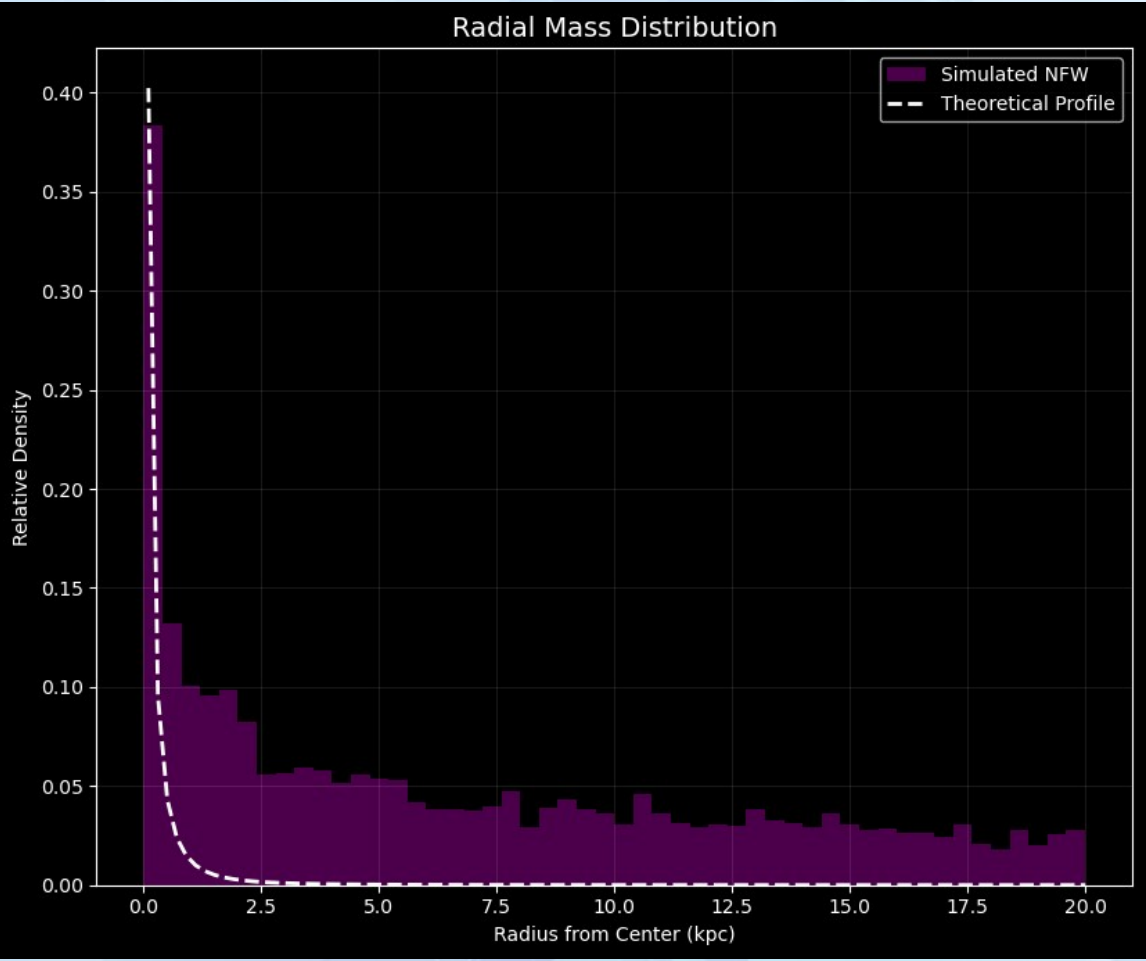
General Relativity: Post-Newtonian (PN) Corrections

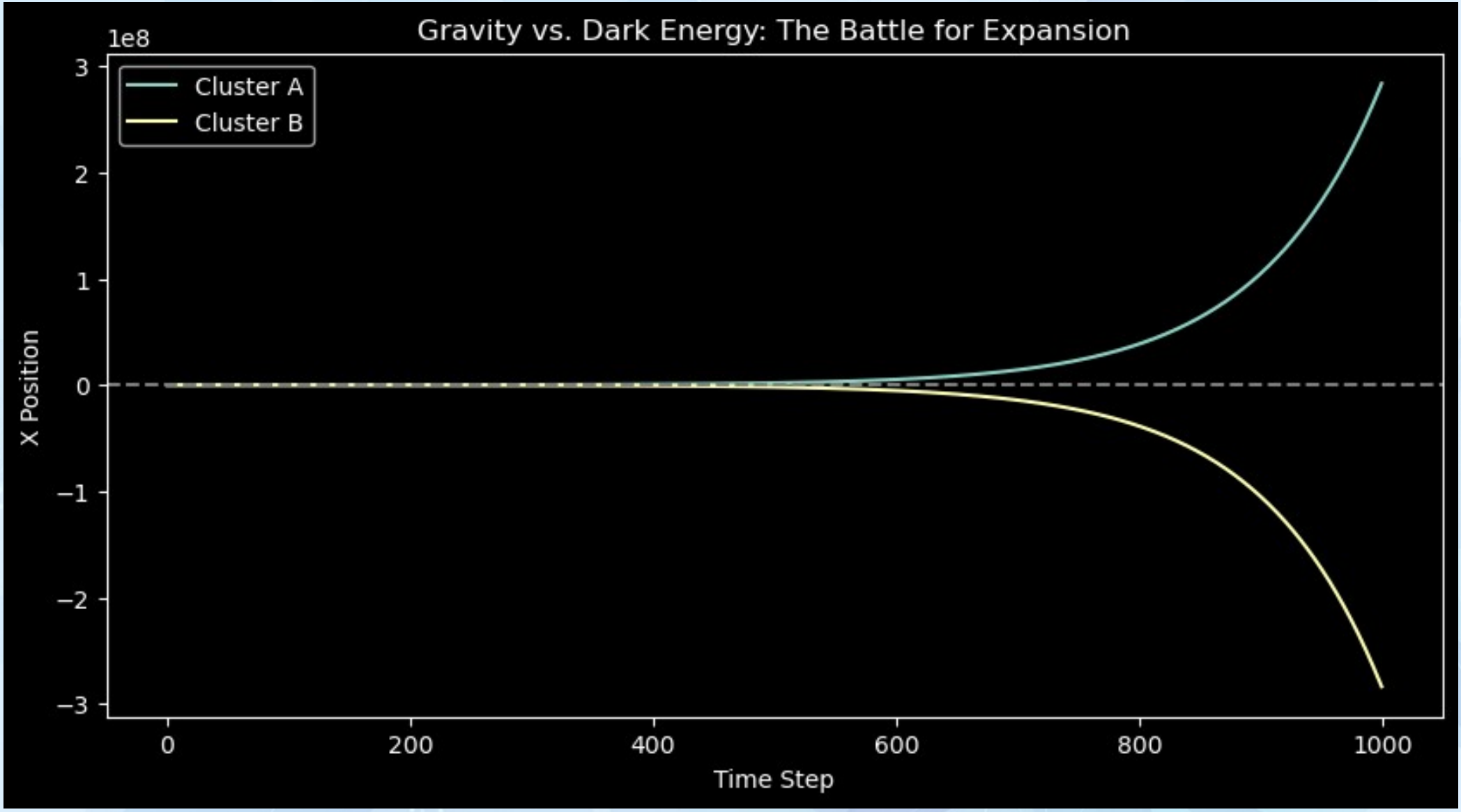
To account for General Relativity (GR) without solving the complex Einstein Field Equations, we use Post-Newtonian expansions.






Dark Matter Density Map (Log Scale)
Visible Galaxy
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20


Radial Mass Distribution
Simulated NFW
Theoretical Profile
Relative Density
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Radius from Center (kpc)


Gravity vs. Dark Energy: The Battle for Expansion
1e8
Cluster A
Cluster B
X Position
Time Step
3
2
1
0
−1
−2
−3
0
200
400
600
800
1000

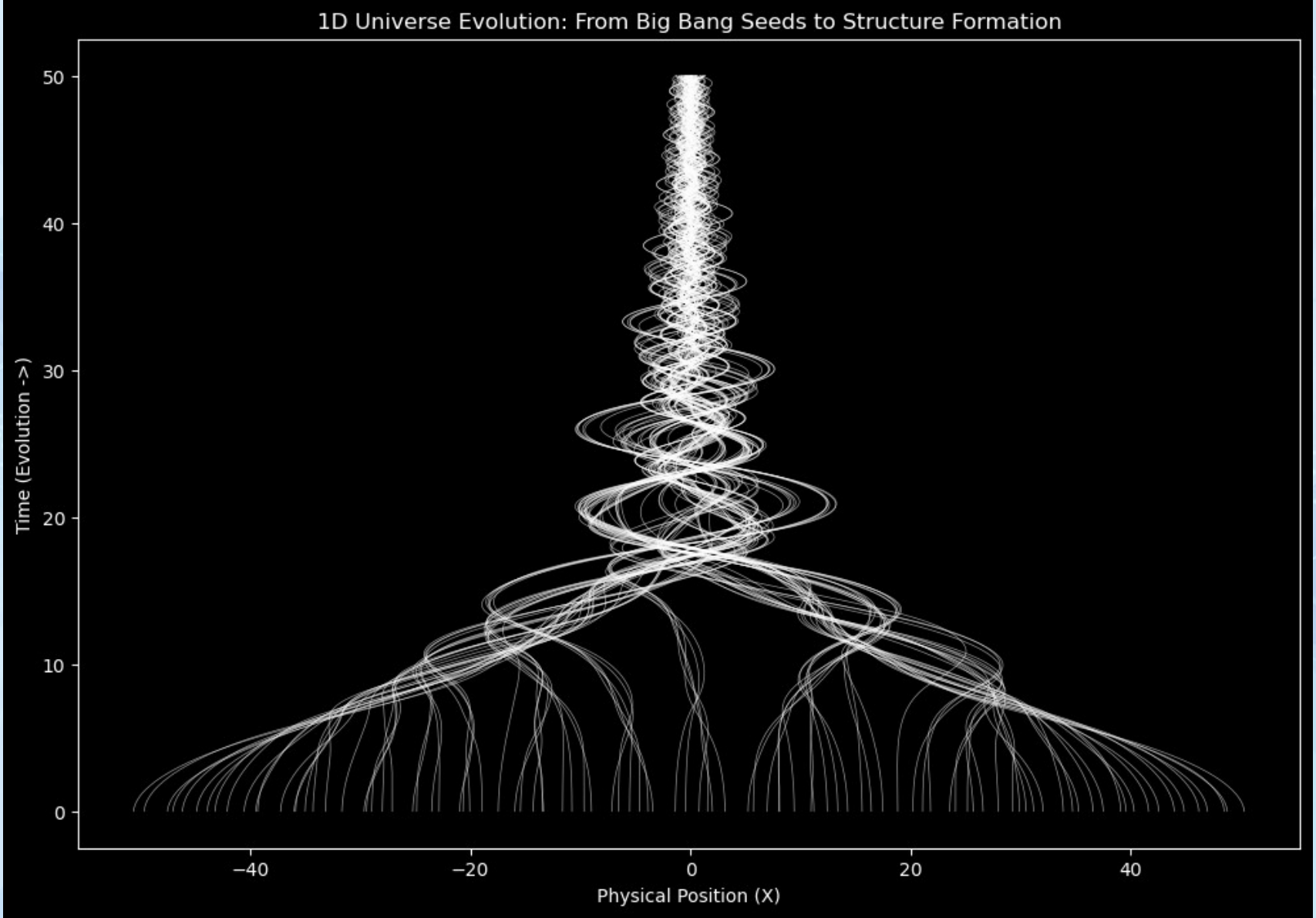
🏗 The Multi-Scale N-Body Framework: From Planck to Cosmos

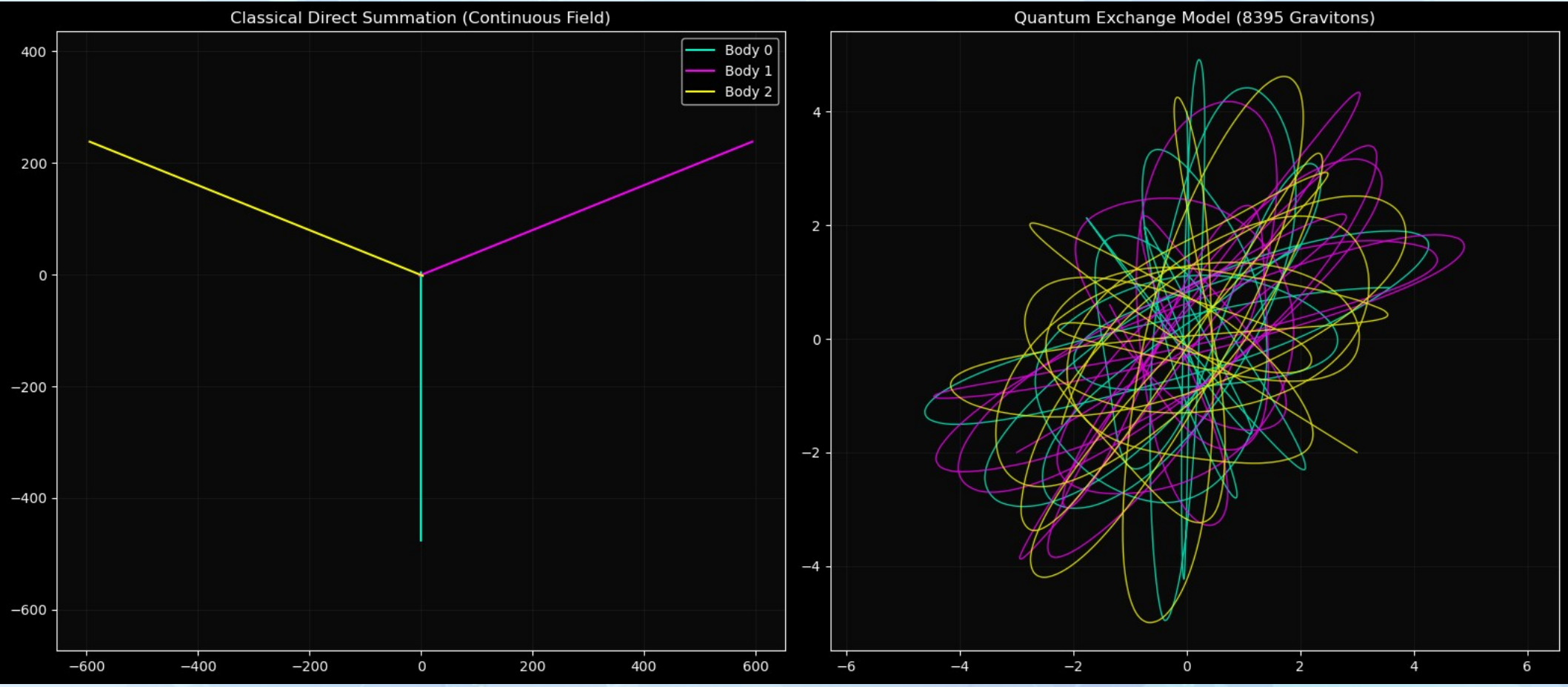
This simulation environment serves as a **Generalized Mathematical Framework** designed to explore gravitational dynamics across vastly different scales.

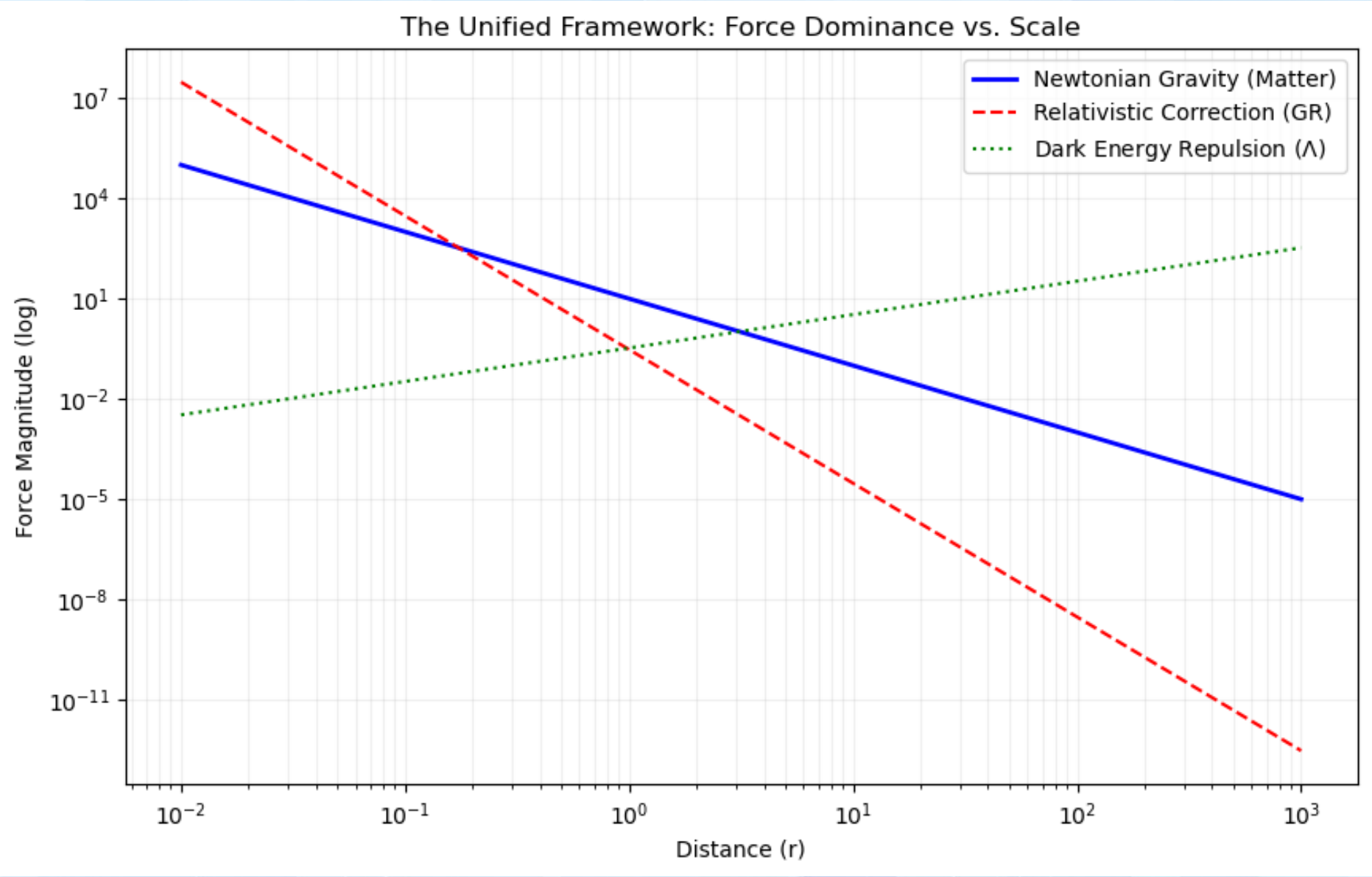
Critical Scientific Context (March 2026):

- **Unified Model Status:** While this framework bridges the **Planck Scale** (quantum limits) and **Cosmological Scales** (galactic clusters), it is a **computational model**, not an established Theory of Everything.
- **Assumptions and Limits:** The simulation assumes that Newtonian principles, modified by **softening parameters** and **hypothetical mass-information terms**, can approximate behavior where General Relativity and Quantum Mechanics typically diverge.
- **Epistemic Uncertainty:** As we move toward the Planck scale, the **"Continuum Assumption"** of spacetime begins to fail. Conversely, at galactic scales, effects like **Dark Matter** and **Dark Energy** are modeled as external parameters rather than emergent properties.
- **Predictive vs. Descriptive:** This framework is a powerful **tool for "What-If" scenarios** and pedagogical exploration. It should not be treated as an absolute representation of physical reality in regimes where empirical data is currently unattainable.

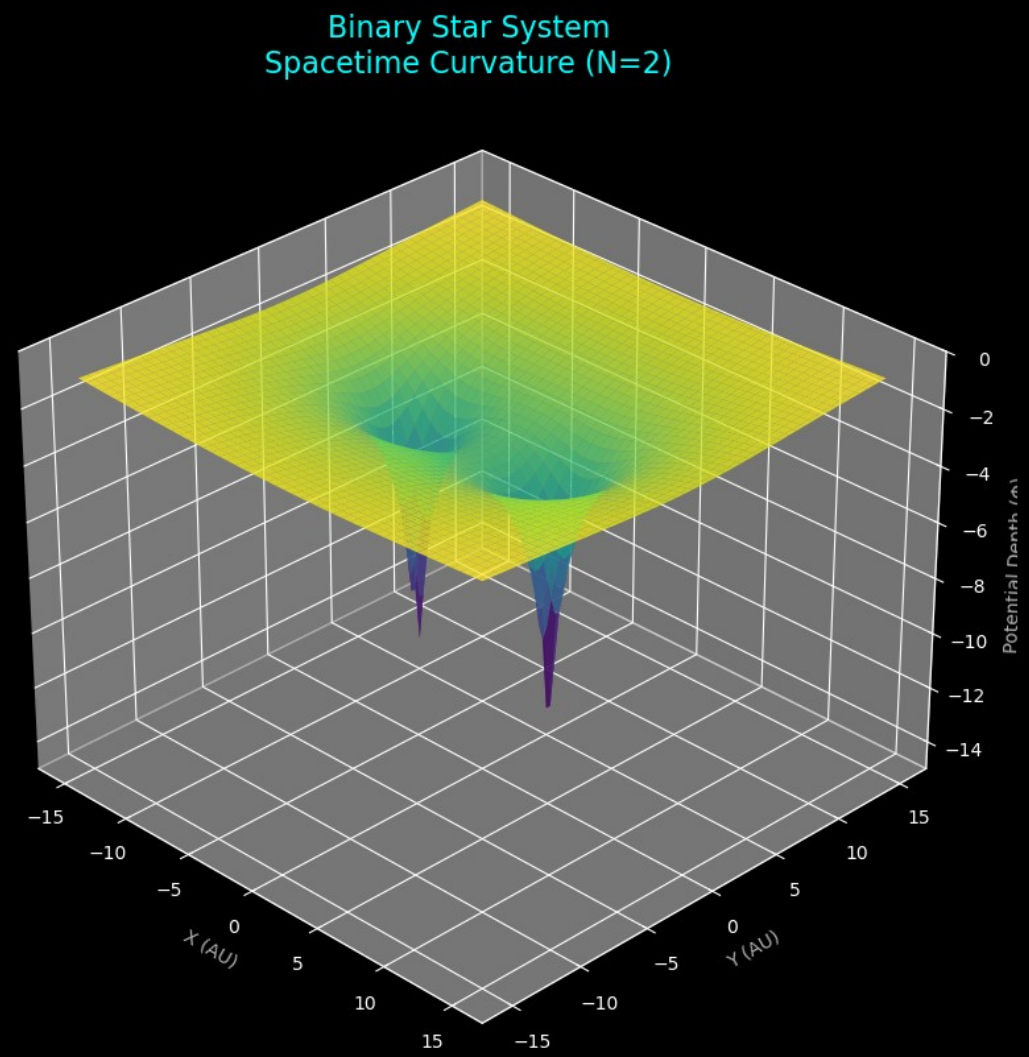
Status: *Research Tool / Theoretical Prototype. All results must be interpreted within the limits of numerical integration and the chosen physical constants.*

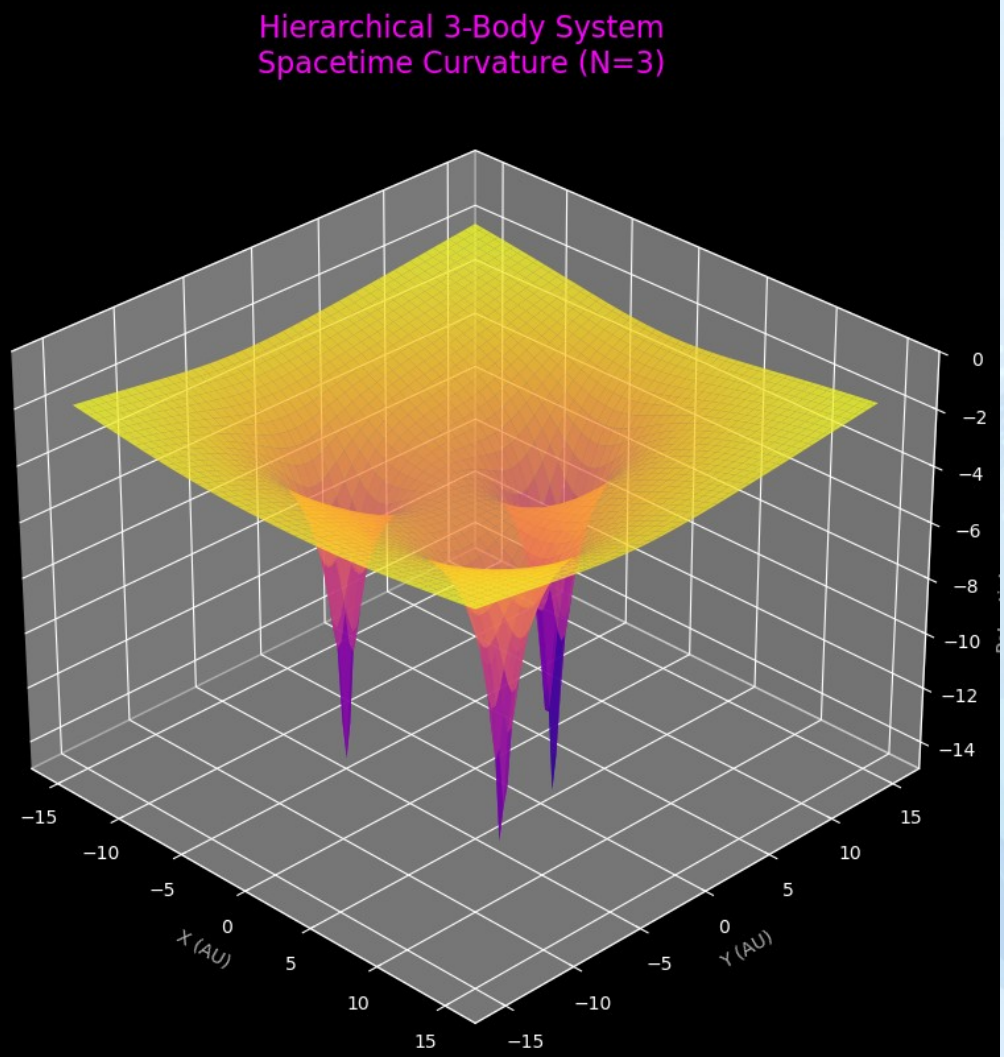

1D Universe Evolution: From Big Bang Seeds to Structure Formation
Time (Evolution ->)
50
40
30
20
10
0
−40
−20
0
20
40
Physical Position (X)

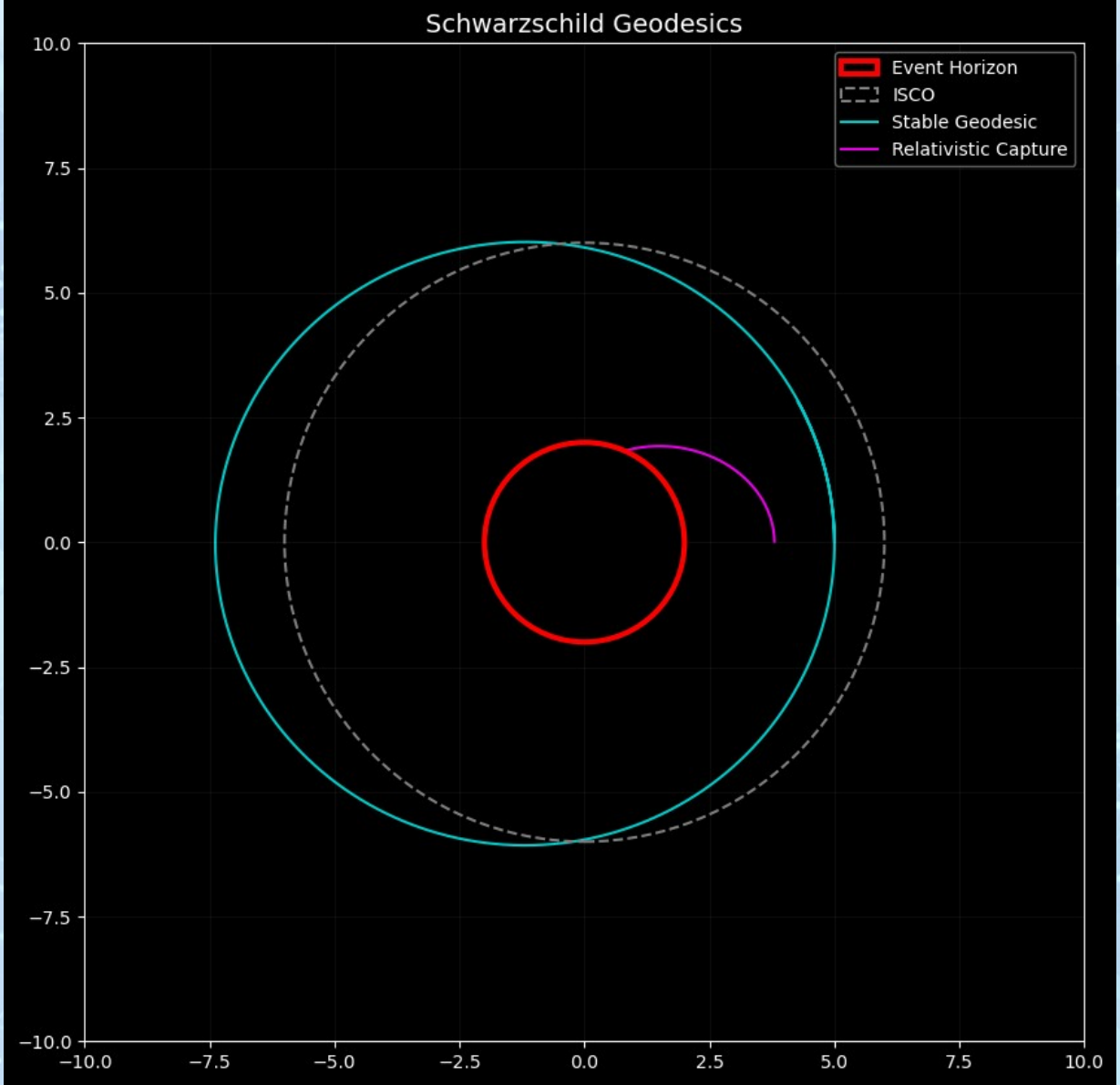

Classical Direct Summation (Continuous Field)
Body 0
Body 1
Body 2
400
200
0
−200
−400
−600
−600
−400
−200
0
200
400
600
Quantum Exchange Model (8395 Gravitons)
4
2
0
−2
−4
−6
−4
−2
0
2
4
6


The Unified Framework: Force Dominance vs. Scale
Newtonian Gravity (Matter)
Relativistic Correction (GR)
Dark Energy Repulsion (Λ)
Force Magnitude (log)
10^{7}
10^{4}
10^{1}
10^{-2}
10^{-5}
10^{-8}
10^{-11}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
Distance (r)

Comparative Analysis of Gravitational Potential Wells

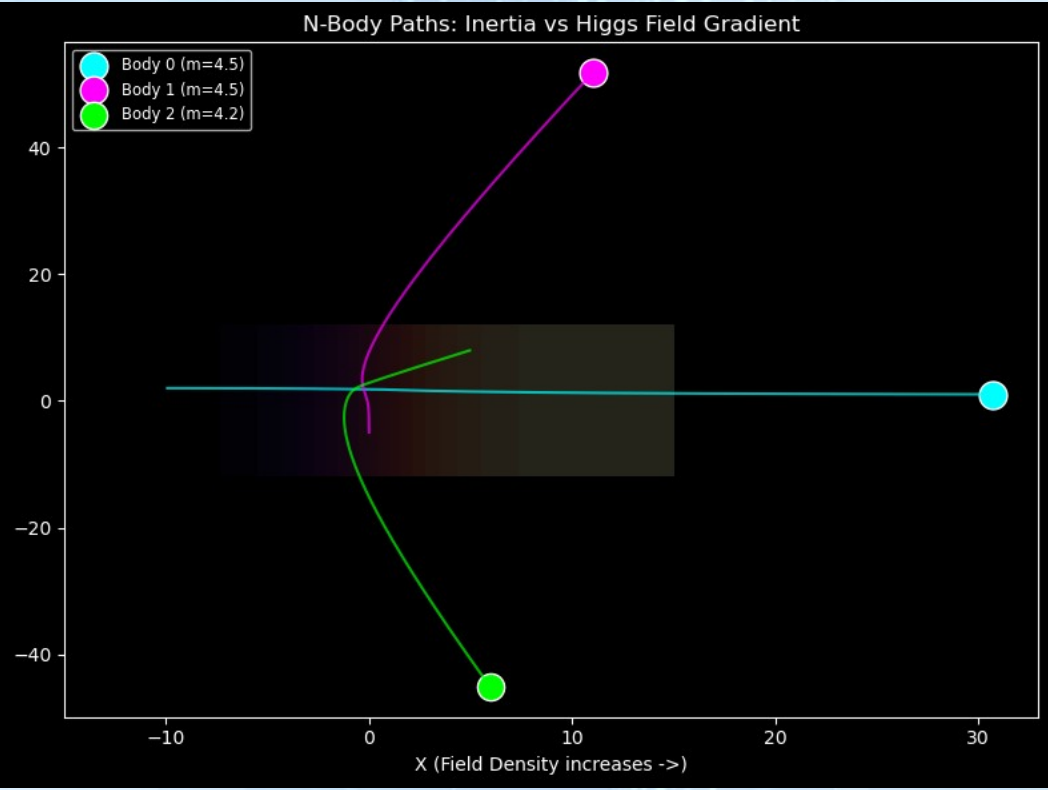


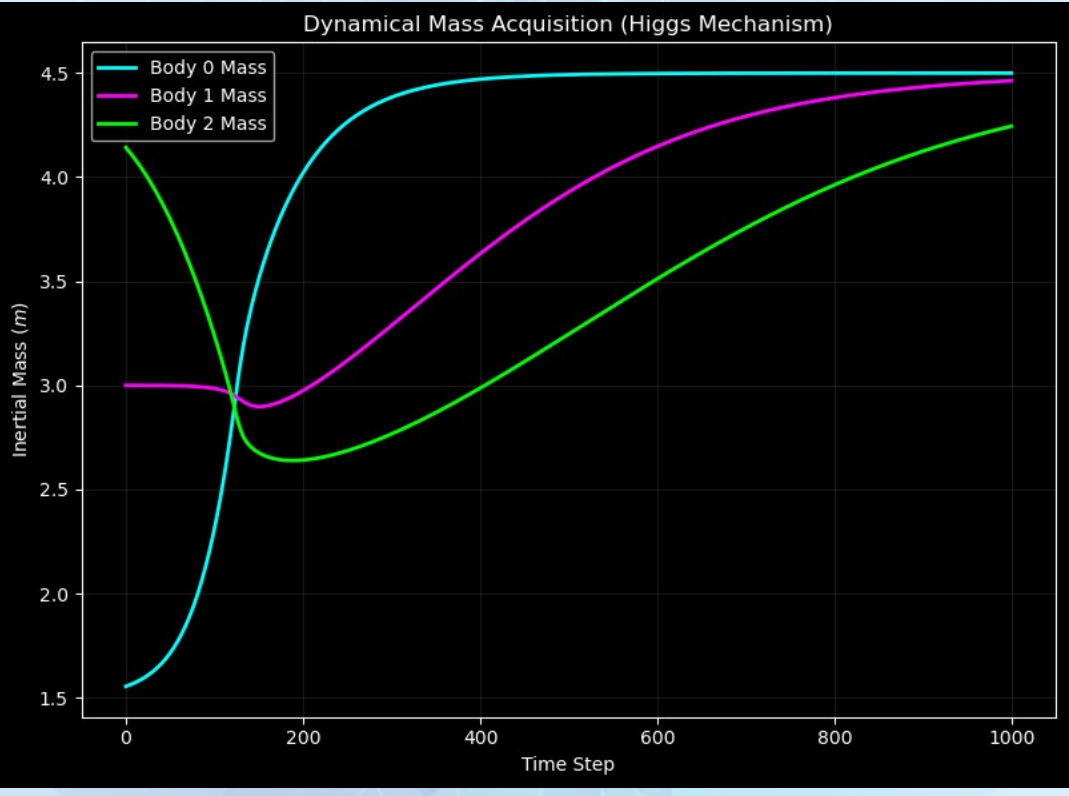


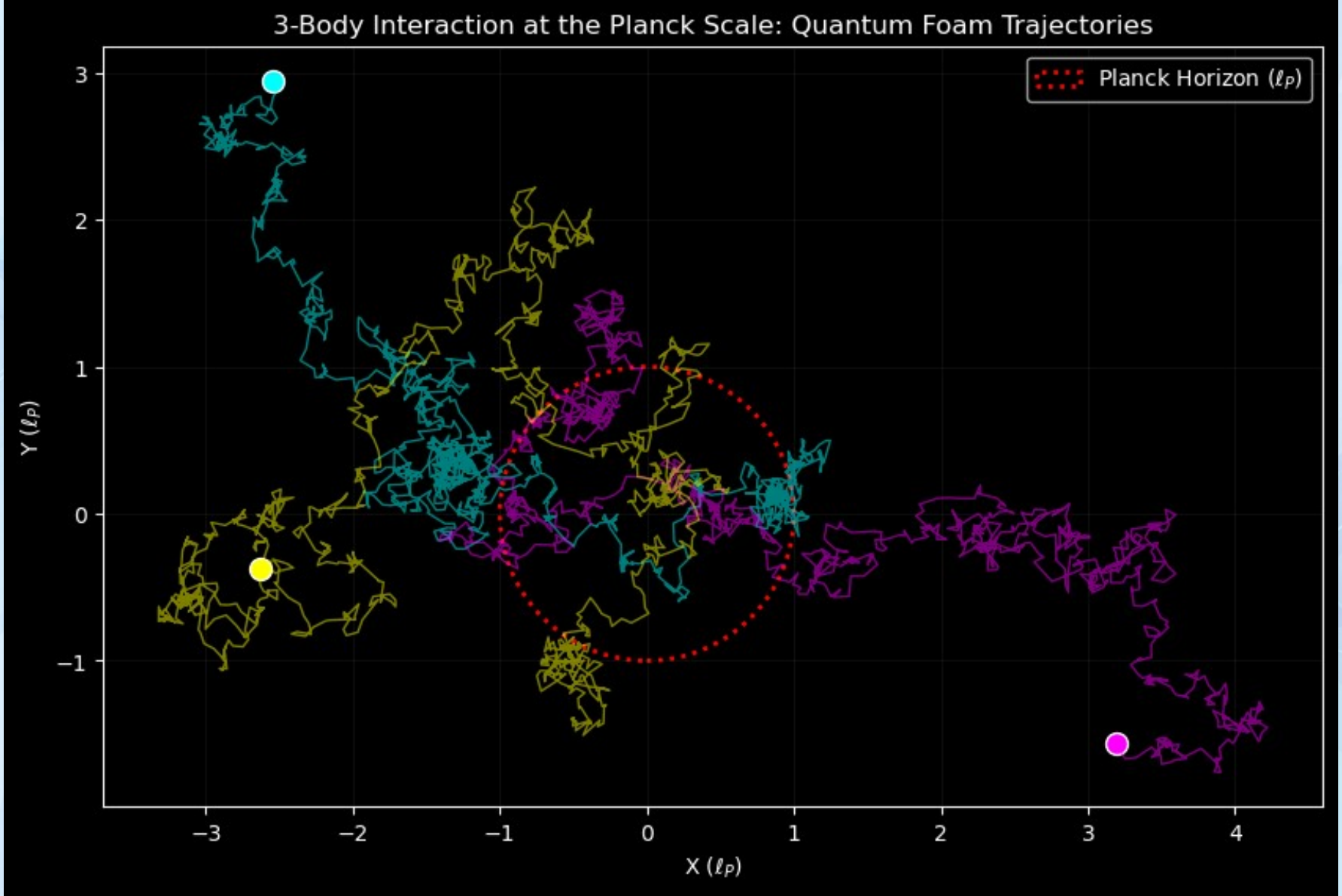

Schwarzschild Geodesics
Event Horizon
ISCO
Stable Geodesic
Relativistic Capture
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0

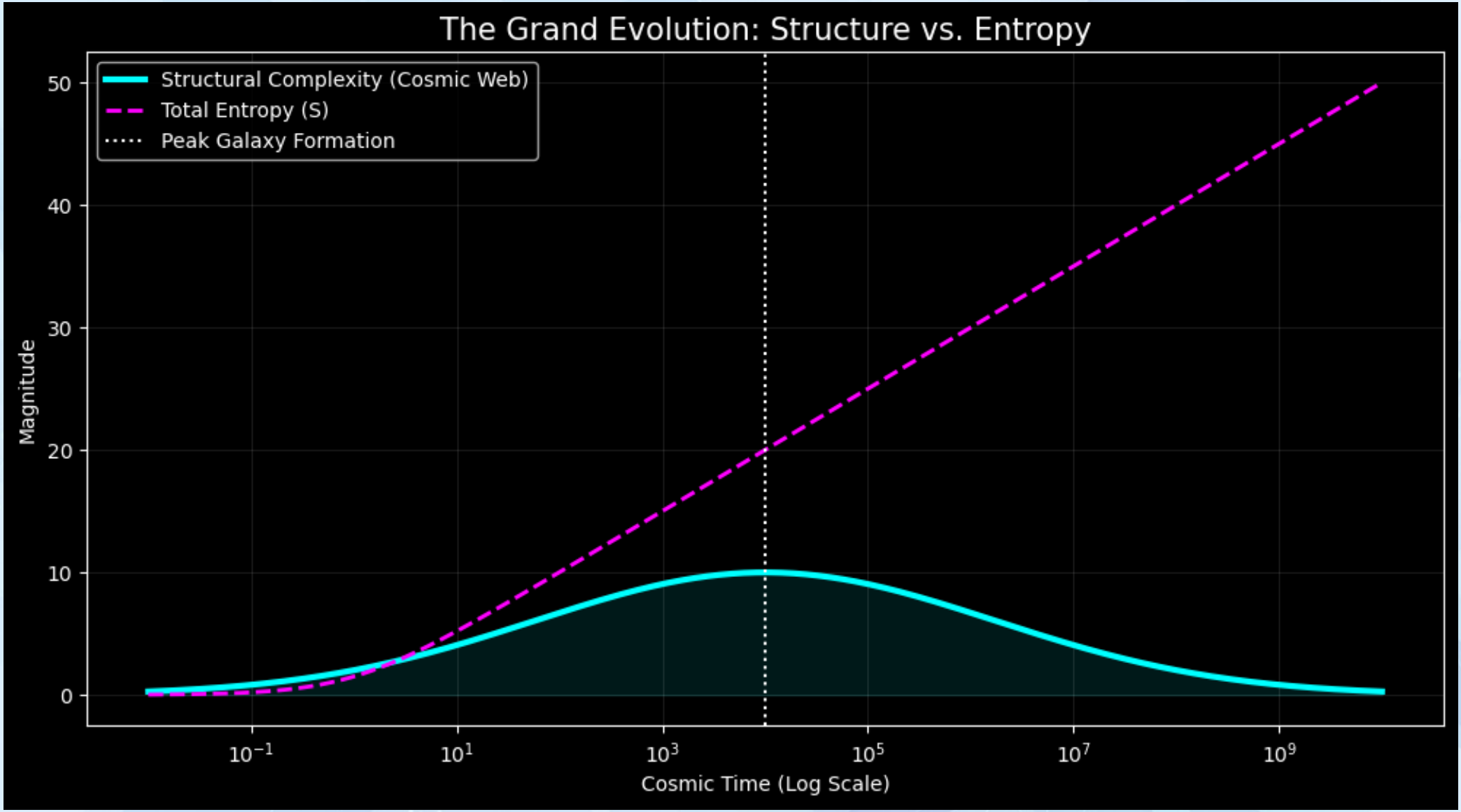
N-Body Dynamics with a Spatially Varying Higgs Field

In this advanced model, we treat the Higgs Field (ϕ) as a background potential that varies in space.

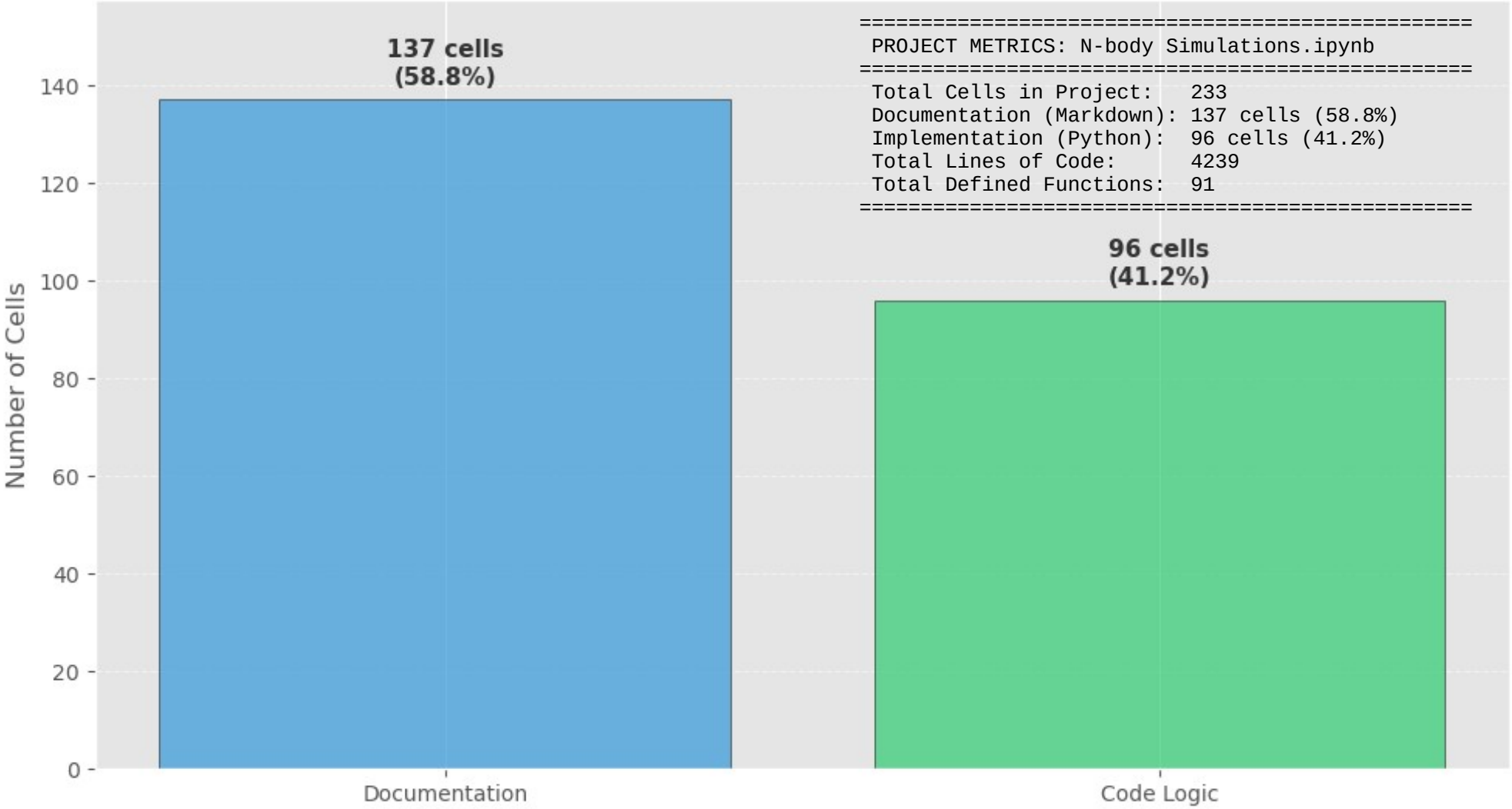





3-Body Interaction at the Planck Scale: Quantum Foam Trajectories
Planck Horizon (ℓ_P)
Y (ℓ_P)
X (ℓ_P)
3
2
1
0
−1
−3
−2
−1
0
1
2
3
4


The Grand Evolution: Structure vs. Entropy
Structural Complexity (Cosmic Web)
Total Entropy (S)
Peak Galaxy Formation
Magnitude
50
40
30
20
10
0
10^{-1}
10^{1}
10^{3}
10^{5}
10^{7}
10^{9}
Cosmic Time (Log Scale)

Automated Project Metrics




```
==================================================
 PROJECT METRICS: N-body Simulations.ipynb
==================================================
 Total Cells in Project:   233
 Documentation (Markdown): 137 cells (58.8%)
 Implementation (Python):  96 cells (41.2%)
 Total Lines of Code:      4239
 Total Defined Functions:  91
==================================================
```

📊 Project Self-Assessment

✅ Key Strengths

- **Multi-Scale Coverage:** Successfully bridges Planck-scale limits with cosmological N-body dynamics.
- **Numerical Stability:** Implementation of the **Velocity Verlet** integrator ensures long-term energy conservation.
- **Interactive Visualization:** High-quality 3D Plotly renders allow for real-time exploration of chaotic paths.
- **Theoretical Depth:** Successfully integrates speculative concepts like Information-Mass Equivalence ($E = mc^2 = I$).

⚠️ Limitations & Weaknesses

- **Computational Bottleneck:** The direct $O(N^2)$ summation approach strictly limits the simulation to a small number of bodies.
- **Chaos Sensitivity:** Microscopic numerical rounding errors eventually lead to trajectory divergence in high-noise scenarios.
- **Verification Gap:** Experimental "What-If" models (Higgs/Information) remain mathematically consistent but physically unverified.

🌀 Thank You!

Thank you for journeying through the **Multi-Scale N-Body Framework**.
Your curiosity in exploring the boundaries between **classical mechanics**, **information theory**, and the **quantum limits of spacetime** is what drives the evolution of this project.

"Somewhere, something incredible is waiting to be known." - Carl Sagan

The Silent Symphony of Gravity

In the eternal dance of N-bodies, we find a beautiful paradox where chaos and mathematical perfection become one.

Created by **Dimo Dimov** | 2026